

Leaky Sanctions^{*}

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October 5, 2026

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Abstract

The sanctions imposed on Russia in 2022 are the most extensive in modern history, yet at least 30% of pre-war EU exports later covered by the bans continued to reach Russia. Combining Russian transaction-level customs records with a new granular dataset on EU export bans, we directly observe evasion and trace its routes through third-country intermediaries. Our estimates imply that the bans raised median effective trade costs from the EU to Russia by about 69%, far below the prohibitive increase under full enforcement. To quantify the consequences, we develop a general-equilibrium trade model with endogenous transshipment. Under observed evasion, Russia's real income falls by 1.26%. Russia's benefit from transshipment is substantial, as fully enforcing the same bans raises this loss by 32%. The model implies that intermediary hubs can earn up to USD billion from facilitating evasion. Enforcement against current routes has limited returns, because closing every route used under observed evasion recovers less than one third of the gap to full enforcement, with trade shifting to previously unused routes. A targeted expansion of the current product list by only 141 products generates a comparable additional loss for Russia at a lower cost to the EU.

Keywords: Evasion, Sanctions, Export Bans, Transshipment, Geoeconomics.

JEL-Classification: F11, F13, F14, F51, F52.

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1 Introduction

Russia’s invasion of Ukraine in February 2022 triggered the most comprehensive sanctions episode in modern history. A broad coalition consisting of the European Union (EU), the United States, and other allies froze assets, severed financial ties, and banned thousands of products from export to Russia, from advanced semiconductors to champagne and luxury cars. The immediate objective of the export bans was clear: to cut off Russian access to Western goods. But whether such bans are successful depends on how easily they can be evaded. This paper measures the extent and organization of sanctions evasion, quantifies how it changes the effects of the export bans, and compares alternative ways of strengthening them.

Measuring evasion is challenging because the underlying activities are illicit and leave little trace in standard trade data. In our setting, however, one important form of evasion is directly visible. Russian transaction-level customs data report both the country of origin and the country of dispatch for each shipment, allowing us to identify EU-origin goods that reach Russia through third countries after the bans take effect. We combine these records with a new product-level dataset that identifies the precise scope and timing of the EU export restrictions.

The EU bans prohibit both direct and indirect exports to Russia. Consistent with this legal framework, direct EU exports of fully banned products fall to zero in official EU trade statistics once the restrictions become binding. Yet Russian customs records continue to report imports of the same products with an EU country of origin. We classify these transactions as evasion and find that their average monthly value remains at least 26% of its 2019 level. Other forms of evasion, including product misclassification or false declarations of origin, are not visible in our data. Our measure should therefore be interpreted as a lower bound.¹

Evasion occurs through two distinct transshipment modes. Under re-exporting, intermediaries in third countries purchase sanctioned goods from EU exporters and ship them onward to Russia, becoming the seller of record while the product’s origin remains unchanged. For example, Ferrari cars, produced exclusively in Italy, were exported directly to Russia before the war but subsequently reached Russia via Belarus or the United Arab Emirates, with Italian origin preserved but the exporting firm shifting from Italian manufacturers to intermediary companies. Under in-transit transshipment, exporters nominally declare a third country as the final destination, such as Kazakhstan or Belarus, while the goods enter Russia and never continue onward. Both modes exploit gaps in the enforcement of export controls: reexporting circumvents the ban by changing who sells,

¹ Transshipment through third countries has been documented in other trade-policy settings; see, for example, *Financial Times*, “Chinese exporters ‘wash’ products in third countries to avoid tariffs,” May 4, 2025, <https://www.ft.com/content/147fddb-7031-4347-9251-4425614e138d>.

while in-transit transshipment circumvents it by misrepresenting the ultimate destination of goods. In-transit transshipment dominates immediately after the bans enter into force in July 2022, reflecting the ease with which destination declarations can be falsified at short notice. As enforcement tightens, with stricter rules curbing transit flows beginning in mid-2023, reexporting rapidly displaces in-transit transshipment as the primary evasion mode. This dynamic suggests that sanctions evasion is not static, with intermediary networks adapting as enforcement closes off particular channels.

Re-exporting is highly concentrated in a small number of hubs. The majority of re-exported EU-origin, export-banned goods pass through just three countries—Türkiye, Belarus, and China—with Türkiye alone accounting for more than half of the total re-exported value. At the same time, re-exporting is not confined to these hubs: nearly one-fifth of re-exported trade is routed through a broader set of intermediary countries. Within this group, activity is itself concentrated, with the United Arab Emirates, Uzbekistan, and Serbia emerging as the most important secondary hubs.

Evasion differs substantially across products. For 29% of fully banned products, average post-ban import values exceed their 2019 level. Across the broader distribution, evasion is higher for products with more alternative suppliers, more pre-war third-country routing, lower shipping weight, and a less concentrated set of Russian buyers. These patterns are consistent with lower setup and concealment costs when goods already move through broad commercial networks. The extreme right tail looks different. Products whose imports rise above their pre-war levels tend to have fewer alternative suppliers and less prior third-country routing. For these goods, the difficulty of substitution appears to dominate: Russia establishes new routes despite the higher cost of doing so.

We find that the export bans created a large but incomplete barrier to trade. To measure how strongly they restrict trade in practice, we estimate a triple-difference gravity specification at the HS6–origin–month level. The design compares fully banned with never-sanctioned products, EU origins and countries mirroring the EU restrictions with non-coalition origins, and periods before and after the bans. Product–origin, origin–month, and product–month fixed effects absorb persistent comparative advantage, origin-wide shocks, and changes in Russian product demand. The event study shows no differential pre-trends and a sharp decline once the initial bans become binding. Affected EU-origin flows fall by roughly 80% relative to the comparison groups. In the calibrated model, the estimate implies that the bans raise median effective trade costs from the EU to Russia by about 69%, far below the prohibitive increase under full enforcement.

To quantify the consequences of this incomplete enforcement, we develop a general-equilibrium trade model with endogenous transshipment. The model distinguishes evasion from the legal adjustment margins that remain available to Russia. After the bans, Russia can switch to non-sanctioning suppliers or other products (standard trade diversion), while

banned coalition-origin goods can continue to reach it through third-country intermediaries (evasion). Enforcement can close the latter channel, but it cannot prevent the former adjustments.

The model is calibrated at the 6-digit HS level. Within each product, Russian buyers can substitute across producer countries. We further group products into military-relevant goods, other critical goods, and all remaining goods, allowing substitution to be harder across these broad uses than among products within one group. This allows Russia to protect expenditure on strategically important goods when their relative prices rise, rather than treating military inputs and ordinary consumer goods as equally close substitutes.

Transshipment depends on both the existing trade network and the incentives of intermediary countries. We use the triangle inequality of trade costs to map pre-war trade costs into the cost of supplying a banned product through each potential hub, building on recent insights from [Foellmi et al. \(2024\)](#). Hubs can differ in their ability to facilitate this circumvention, and the most competitive hub can charge a premium over its rerouting cost. Competition from the next-best route limits this premium. The model therefore captures both the additional cost of evasion and the rents that make facilitating it attractive.

We calibrate the pre-sanction economy using 2019 trade and production data, while the post-sanction evidence then disciplines the transshipment technology in the post-sanction world. The customs routes determines the hubs' comparative advantage, telling us where evasion takes place, while the gravity estimate disciplines the overall cost of rerouting. The model then determines the selected routes, hub premia, and the resulting changes in trade and income. The pre-war network already identifies most of the main hubs, and the calibrated model also improves the fit of routing patterns that are not directly targeted in the calibration.

We find that transshipment substantially weakens the export bans. Under observed evasion, Russia's real income falls by 1.26% relative to the no-sanctions baseline. With no transshipment, the same product restrictions raise the loss to 1.66%, an increase of 32% relative to the observed-evasion loss. Equivalently, evasion offsets 24% of the loss under full enforcement. We refer to the absolute difference between the two losses as the *enforcement gap*.

The decline in targeted trade overstates the decline in the goods available to Russia. Imports of banned coalition-origin products fall by 76% under observed evasion, but total imports fall by only 9.1% as Russia switches suppliers and reallocates expenditure across products. The model's measure of access to military-relevant goods falls by 3.6% under observed evasion, compared with 4.0% under full enforcement. Thus, substitution substantially limits the effect of the bans beyond the targeted bilateral flows.

Hub activity is highly concentrated, and evasion creates gains for a small number of third countries. Under our benchmark markup assumption, intermediary hubs earn up to USD billion from rerouted trade. Türkiye and the United Arab Emirates gain mainly through these rents, while China benefits primarily from expanding its own exports to Russia. The losses within the EU are much smaller on average, but concentrated among members with stronger pre-war exposure to Russia. Under observed evasion, the median member state loses 0.036% of real income.

Enforcement aimed at current routes has limited returns. One natural response, reflected in current sanctions enforcement, is to target the firms and intermediaries carrying banned goods along active routes (Section 7.1 discusses recent U.S. and EU enforcement measures of this kind). Even if every route used under observed transshipment were identified and closed, only 30% of the enforcement gap would be recovered. The reason is the equilibrium response of the transshipment network, because once the active routes are removed, previously unused routes become the most attractive alternatives.

Removing an entire country as a hub goes further because it also blocks routes that are not initially used. If the country even joins the coalition, it additionally restricts its own exports to Russia. The importance of these two channels differs remarkably across countries. For Türkiye and the United Arab Emirates, most of the accession effect comes from ending their role as intermediaries. For China, restricting Chinese-origin exports matters much more than removing its hub role. For Türkiye, we also compare the cost of joining while retaining access to the EU–Türkiye Customs Union with the cost of remaining outside after that relationship is downgraded to a free trade agreement. The latter is substantially more costly to Türkiye, suggesting that the Customs Union provides meaningful leverage, although a downgrade would also be costly to the EU.

We show that expanding the scope of products covered by the export bans instead can be very effective. Among the policies we compare, a targeted expansion of the product list (keeping the transshipment technology fixed) reaches the route-closure benchmark at the lowest cost to the EU. We rank candidate products by the loss they impose on Russia per dollar of newly covered exports, taking the available rerouting options into account, and then evaluate the resulting packages in general equilibrium. Adding only 141 products, about 4% of the candidate products and a 4% increase in the baseline export value covered by the current list, generates approximately the same additional loss for Russia as closing every route used under observed evasion. Expanding the list by 965 products, equivalent to a 25% increase in covered export value, restores the no-transshipment loss under the current restrictions. Product choice matters as well, because at the exposure budget of the smaller package, a ranking that ignores transshipment generates 22% less additional damage.

Relation to the literature. We contribute to several strands of the literature. First, we contribute to work on the circumvention of trade barriers. A large literature studies evasion in the context of taxes (Fisman and Wei, 2004; Laffitte and Toubal, 2022) and tariffs (Fisman et al., 2008; Javorcik and Narciso, 2008), with renewed interest following the U.S.–China trade war (Che et al., 2025; Freund, 2025; Iyoha et al., 2024) and the recent sanctions against Russia (Chupilkin et al., 2025, 2023b; Scheckenhofer et al., 2025; Tyazhelnikov and Romalis, 2024). We observe an important form of evasion directly because the Russian customs records report the origin and dispatch country of each shipment separately. We then use the observed routes to discipline a general-equilibrium model in which transshipment responds endogenously to enforcement and generates income for intermediary countries. This allows us to quantify the aggregate consequences of evasion, distinguish it from legal trade diversion, and study how trade moves to new routes when existing ones are closed.

Second, we contribute to the growing literature on the trade and income effects of economic sanctions, particularly those imposed on Russia. Existing work studies these effects using firm-level data (Corsetti et al., 2025; Crozet and Hinz, 2020; Crozet et al., 2021; Egorov et al., 2025), more aggregated trade data (Egorov et al., 2025), and quantitative general-equilibrium trade models (Chowdhry et al., 2024; Flach et al., 2024; Hausmann et al., 2024; Hinz and Monastyrchenko, 2022). Existing quantitative analyses typically evaluate the announced restrictions under full enforcement, while reduced-form evidence on circumvention does not by itself recover the aggregate consequences of the remaining trade. We bridge these approaches by estimating the effective severity of the export bans and embedding the estimate in a model of endogenous transshipment. This allows us to compare observed and full enforcement, measure the incidence of evasion across Russia, coalition members, and intermediary countries, and evaluate alternative ways of strengthening the restrictions.

Third, we contribute to the measurement of trade sanctions. We compile a uniquely detailed, tariff line-level dataset on EU export bans that identifies which products are affected, the precise timing of each restriction, and whether bans apply fully or only to specific technical characteristics. Constructing such data is non-trivial because the EU’s sanctions regime varies widely across products and over time, often within narrowly defined tariff lines. For example, within a single tariff line covering industrial pumps, only certain models—such as those suitable for chemical-weapons production—are restricted, while others remain exportable. As a result, more than half of tariff lines are initially only partially restricted rather than fully banned.² Existing sanctions datasets either code sanctions at the country-pair level, with the goal of covering all sanctions worldwide

² Partial sanctions reflect the EU’s “smart sanctions” approach, which aims to restrict only items with sensitive or military applications while minimizing collateral damage to legitimate trade.

and across all sanction types, not only trade sanctions (Felbermayr et al., 2024; Morgan et al., 2023), or include product-level information without resolving the precise legal scope of restrictions within tariff lines, as in commonly used sources such as the Global Trade Alert or EU Council regulations (Crozet and Hinz, 2020; Flach et al., 2024). As a consequence, they would misclassify a large volume of legal shipments as illicit and mechanically overstate evasion. We overcome this limitation by identifying, for each CN8 product, whether restrictions apply fully or only to specific technical specifications, as well as the exact date at which they become binding, based on TARIC and a comprehensive review of the European Commission’s legal gazettes.

Finally, we show how to bring a multi-product quantitative trade model to highly disaggregated data (see Caliendo and Parro, 2022; Costinot and Rodríguez-Clare, 2014, for overviews). We quantify an Eaton and Kortum (2002)-type framework at the HS6 level despite the absence of globally comparable domestic expenditure at this level of detail. This granularity allows product-specific trade costs and elasticities, distinguishes military-relevant and other critical goods from ordinary products, and maps the legal product coverage of the export bans directly into the model. We further incorporate transshipment through competing hubs and the income they earn from intermediating banned trade.

The paper proceeds as follows. Section 2 introduces the customs and export-ban data and explains how we identify evasion. Section 3 documents its magnitude, transshipment modes, hubs, and variation across products. Section 5 develops the quantitative model. Section 4 estimates the effective trade-cost increase generated by the export bans and provides the empirical moment used to calibrate the overall cost of transshipment. Section 6 quantifies the effects of observed evasion and full enforcement, and Section 7 compares alternative ways of strengthening the bans. Section 8 concludes.

2 Measuring Evasion

The full-scale Russian invasion of Ukraine in February 2022 marked the largest military conflict in Europe since World War II. In response, Western allies, led by the U.S. and the EU, imposed an extensive and evolving sanctions regime against Russia, spanning multiple policy domains including finance (e.g., exclusion from SWIFT and asset freezes), trade (import and export bans), transport (airspace restrictions), energy (oil and gas), and targeted sanctions on firms and individuals. In this paper, we focus on EU export bans, which form a central pillar of the sanctions regime and are particularly well suited

for studying evasion.³ The scope of the export bans expanded substantially over time as successive sanctions packages were adopted, ultimately covering a broad range of products (Table A.1).

Because the bans apply to both direct and indirect exports, EU firms face strong incentives to comply formally when exporting banned products to third countries: misreporting origin or misclassifying products would constitute a direct sanctions violation. Russia does not enforce EU export bans, so Russian importers face no enforcement-related incentives to misreport the country of origin in customs declarations. Customs authorities therefore continue to record the declared country of origin for imports, including shipments that enter Russia via third countries. As a result, goods that enter Russia despite the export bans often retain their EU origin in customs records, making transshipment and hence evasion directly observable in the data.⁴ Our analysis focuses on this observable margin of evasion. To the extent that evasion occurs through relabeling of origin or product codes by intermediary firms, it is not directly observed in the data, and our estimates should therefore be interpreted as a lower bound. This feature distinguishes export bans enforced at origin from import tariffs or import bans, where evasion operates through misreporting at the border and is unobservable in customs data.

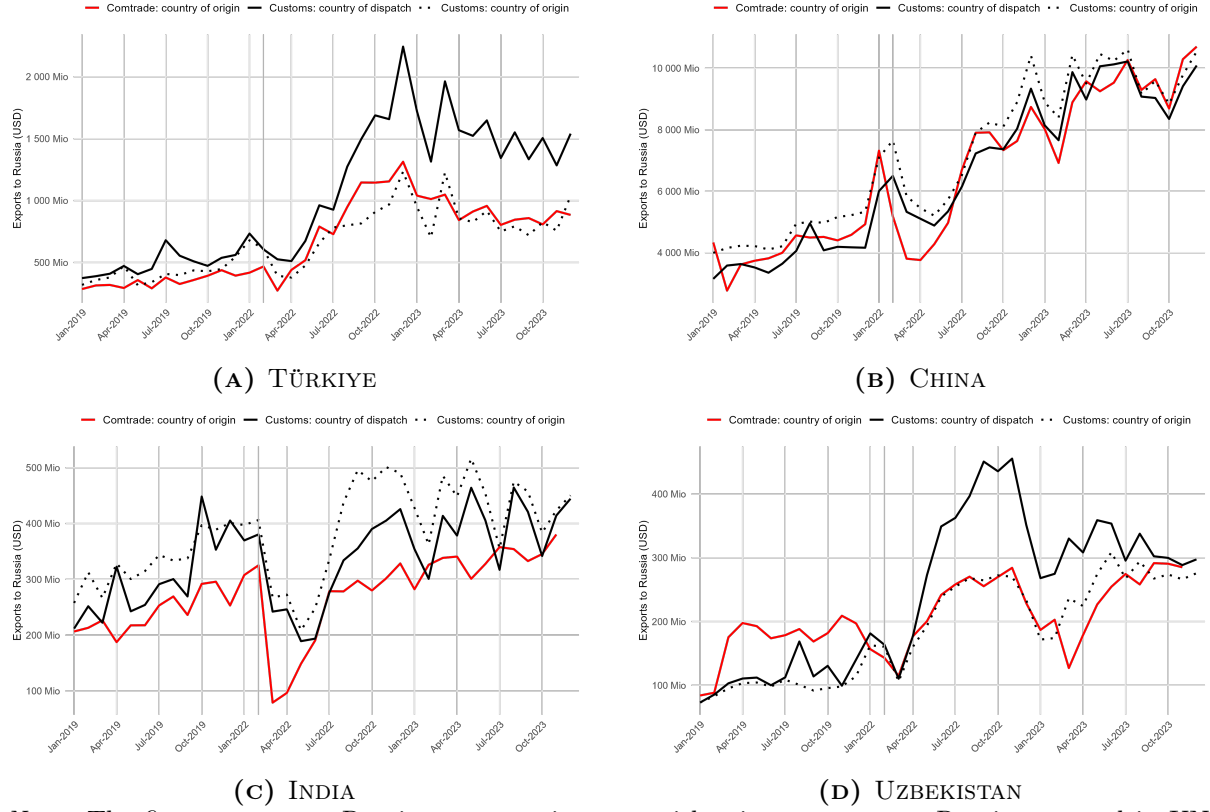
2.1 Russian Customs Trade Data

To observe evasion through transshipment that does not involve manipulation of the true origin, we use confidential Russian transaction-level customs data covering the universe of import declarations. These data report firm-to-firm transactions and include detailed information on product descriptions, 10-digit product codes, quantities, values, and, crucially, the declared country of origin for each shipment. Furthermore, the data distinguish between the country of origin—where the product was produced—and the country of dispatch—the location from which the product was last shipped to Russia. Prior to the war, these customs records formed the basis for Russia’s official aggregate trade statistics reported through UN Comtrade and domestic statistical releases. Other studies using these data show that they align closely with other trade statistics (Babina et al., 2026; Chupilkin et al., 2023a; Egorov et al., 2025; Korovkin and Makarin, 2023).

³ We refer to both outright export prohibitions and export licensing requirements as *export bans*. Although the EU’s sanctions regime distinguishes between the two, licenses for restricted exports to Russia are rarely granted outside narrowly defined exemptions (e.g., humanitarian uses). In practice, licensing requirements therefore function as bans.

⁴ Under Russia’s parallel-importing regime, imports of branded goods are permitted without the authorization of the trademark holder, including goods originating in sanctioning countries. Russian customs authorities therefore continue to accept and record the original country of origin for such shipments.

A natural concern is whether Russian customs data remain reliable in the post-invasion period. Systematic manipulation of customs microdata is considerably more difficult than manipulation of aggregate statistics such as GDP or headline trade balances. In 2023 alone, the customs records contain roughly 30 million individual transactions, which would have to be altered consistently across products, firms, origins, and time.



Note: The figure compares Russian customs imports with mirror exports to Russia reported in UN Comtrade, by month, year, and partner country. Data are shown for 2019 and for 2022–2023 for the largest exporters to Russia outside the sanctioning coalition, for which UN Comtrade data are available. Trade values are expressed in millions of U.S. dollars.

FIGURE 1. VALIDATION OF RUSSIAN CUSTOMS DATA

We further assess data quality by comparing Russian import records to mirror export data from UN Comtrade, which report exports to Russia by country of origin as declared by partner countries. Such comparisons are uninformative for sanctioning countries, for which discrepancies are expected by construction and are precisely how we measure evasion. We therefore focus on non-sanctioning countries, whose exports to Russia remain legal and unrestricted and for which there is no incentive to misreport trade. If Russian customs data were systematically manipulated, imports from these countries would also diverge from partner-reported exports. We examine the four largest post-invasion exporters to Russia outside the sanctioning coalition for which UN Comtrade data are available; additional examples are reported in Appendix Figure A.2.

Figure 1 shows that, for Türkiye, China, and Uzbekistan, UN Comtrade mirror exports closely track Russian customs imports by country of origin. For India, Comtrade flows are systematically lower, but this pattern predates the full-scale invasion of Ukraine in February 2022 and likely reflects known limitations of mirror statistics, such as valuation differences between CIF and FOB prices or reporting thresholds, rather than manipulation of the customs data. Overall, these comparisons suggest that the Russian customs data reliably capture trade flows and origin information.

2.2 Data on EU Export Bans

Information on origin alone is not sufficient to measure evasion; doing so also requires knowing which EU products are legally subject to export bans and when these restrictions take effect. We therefore compile a detailed dataset on EU export bans that identifies the affected products and the timing of the restrictions. Constructing such a dataset is non-trivial because export bans are often defined at a level of technical detail finer than the CN8 classification used in EU trade statistics and therefore apply only to specific product variants within a tariff line. For example, within a given CN8 category of industrial pumps, only variants with technical characteristics suitable for chemical weapons production may be restricted, while otherwise similar pumps remain legally exportable. This reflects the logic of “smart sanctions”, which aim to restrict access to strategically sensitive inputs while limiting collateral losses for EU exporters. If legally exportable product variants are mistakenly classified as banned, any continued exports of these products would be incorrectly counted as evasion. Accurate identification of the legal scope of export bans is therefore essential for our analysis.

We combine information from EU Council regulations with the EU’s Integrated Tariff (TARIC) to recover all trade restrictions affecting EU–Russia trade and their type. TARIC records all EU-measures governing the import and export of goods, including tariffs, documentation requirements, and import and export restrictions.⁵ The database is updated daily and covers all CN8 product codes—over 10,000 products—since 1994. Using TARIC, we retrieve all export restrictions affecting trade with Russia at a daily frequency. When sanctions are defined at a more aggregated level, for example, the February 2022 ban on airplane exports under the two-digit product code 88, all CN8 codes within that category inherit the restriction.

TARIC does not distinguish between partial and full export bans; rather, it indicates whether a given product is subject to one or more export restrictions at a given point in time. We distinguish between fully and partially banned products by combining TARIC

⁵ https://taxation-customs.ec.europa.eu/customs-4/calculation-customs-duties/customs-tariff/eu-customs-tariff-taric_en

with a manual review of the relevant EU Council regulations and their annexes. Products are classified as partially banned for two reasons. First, EU Council regulations may specify *ex* CN codes, which restrict only specific product variants within a CN8 category. Second, products covered exclusively by the EU’s dual-use regulation are restricted only for variants meeting specific technical characteristics and are therefore classified as partially banned. Importantly, some products covered by the dual-use regulation are also subject to additional export restrictions. In these cases, all variants within the CN8 category are restricted, and the product is classified as fully banned. TARIC allows us to identify such overlap across legal bases and to distinguish within the set of dual-use products those that are partially versus fully banned.

This distinction is most relevant early in the sanctions regime, when relatively few products were fully banned. Following the second sanctions package of February 25, 2022, export restrictions applied primarily to dual-use goods, aviation products, and oil-refining technologies, leaving many products only partially covered. As subsequent sanctions packages were adopted, the scope of full export bans expanded, and several products initially classified as partially banned became fully banned as entire CN8 categories were explicitly banned. We track these changes over time to maintain a consistent product-level classification.

Furthermore, the correct entry-into-force date can only be retrieved through TARIC. A common approach in the literature, relying on the published EU Council regulations and their annexes, uses the regulation’s publication date as the entry-into-force date for export bans (e.g., [Crozet and Hinz, 2020](#); [Flach et al., 2024](#)). However, this is often inaccurate, as entry-into-force dates do not necessarily coincide with the publication date of the regulation. For example, a regulation published on February 25 may ban the export of oil-refining technology, yet continue to allow exports under contracts concluded before that date until May 27. Such phased implementation and exemptions are difficult to capture when relying only on the legal texts. In contrast, TARIC records unambiguous, product-specific entry-into-force dates.

To combine the sanctions data with the Russian customs data, we aggregate products to the six-digit Harmonized System (HS6) level. An HS6 product is classified as fully sanctioned if all underlying CN8 products are sanctioned, and as partially banned otherwise. We express all products in the 2017 revision of the HS nomenclature (HS2017).⁶ At the monthly frequency, a product is coded as affected by the export bans only if the restriction is in force for at least 15 days of the month.

⁶ We use standard concordance tables provided by the United Nations: <https://unstats.un.org/unsd/classifications/Econ>. HS2017 products can map to multiple HS2022 codes, in which case the HS2017 product is classified as partially sanctioned if some underlying HS2022 codes are not fully banned. This coding choice avoids mechanically overstating the scope of export bans.

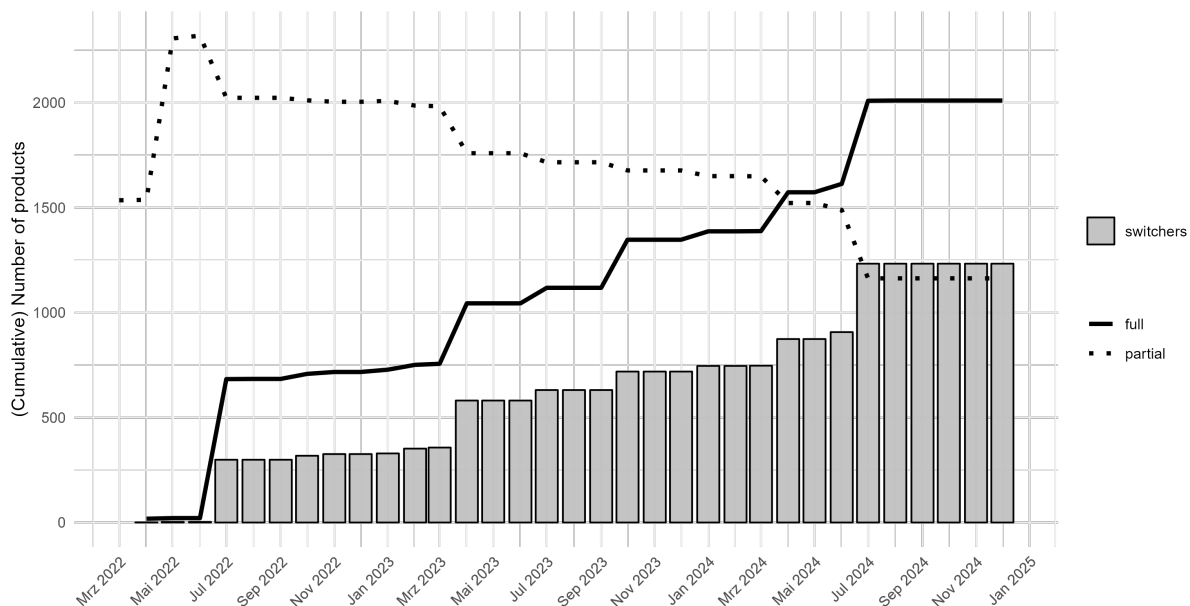


FIGURE 2. EU EXPORT BANS OVER TIME

Note: The figure shows the cumulative number of HS6 products covered by EU export bans over time. The solid line denotes fully banned products, the dotted line partially banned products (accounting for transitions from partial to full coverage), and the gray bars indicate the number of products that switch from partial to full coverage.

By July 2022, 51% of all 6-digit products were subject to EU export bans, driven predominantly by partially sanctioned products (partial: 2,023 products; full: 688 products). By the end of 2024, this pattern had reversed: 38% of all products were fully sanctioned, while only 23% remained partially sanctioned (partial: 1,163 products; full: 2,010 products). Figure 2 illustrates that this shift reflects both the expansion of the sanction regime and the progressive tightening of existing sanctions. In particular, the EU increasingly targeted the remaining unsanctioned CN8 codes within affected 6-digit product categories, effectively converting roughly 1,200 products from partial to full sanction status over the period of almost three years.

Figure A.3 shows that the transition from partial to full export bans is concentrated in a subset of sectors. The shift is particularly pronounced in sectors where restrictions were initially defined through textual provisions, most notably dual-use and high-tech goods in Optics and in Arms & Ammunition. A similar evolution is observed in sectors where sanctions first applied narrowly to specific CN8 products rather than to entire two- or four-digit categories, such as Machinery & Electrical Equipment, and Plastics. By contrast, sectors such as Jewelry and Works of Art exhibit little change in the share of partially sanctioned products over time. In these categories, product-level conditional exemptions for luxury goods persisted, limiting the scope of full export bans.

EU export bans have a sharp and immediate effect on direct trade. As shown in Figure 3, once products become fully subject to export bans, direct EU exports to Russia reported in official EU trade statistics (Comext) drop to zero. By contrast, exports of partially banned products also decline but do not collapse, consistent with these goods remaining largely legally exportable. Exports of non-banned products also decline following the invasion, indicating that the contraction in EU–Russia trade is not driven by export bans alone: Broader sanctions, heightened uncertainty, the retrenchment of multinational firms driven largely by reputational concerns, and weakening Russian demand all contribute to a generalized reduction in direct trade.

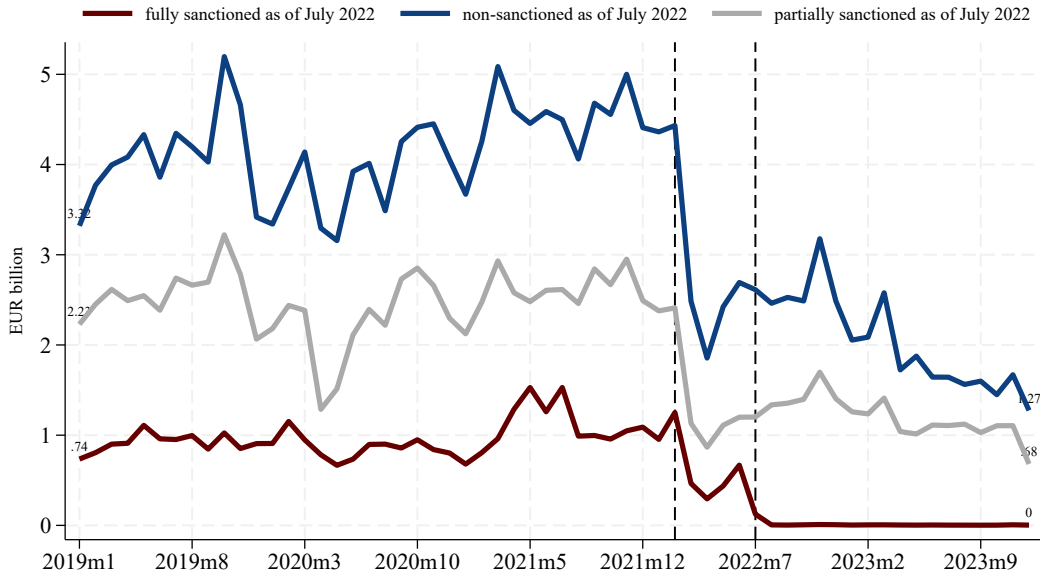


FIGURE 3. DIRECT EU EXPORTS TO RUSSIA

Note: The figure shows monthly EU exports to Russia by export-ban status using official EU trade statistics (Comext), with values in billion euros. Products are classified based on their legal status as of July 2022 into fully banned, partially banned, and non-banned categories.

2.3 Identifying Evasion in Customs Data

We define evasion as the continued import into Russia of products that are fully subject to EU export bans after the bans take effect. A transaction is classified as evasion if it involves a fully banned product, lists an EU member state as the country of origin in Russian customs data, and occurs in a month in which direct EU exports of that product to Russia are prohibited. Because direct EU exports of fully banned products drop to zero after the bans enter into force, any such imports must occur indirectly. Unlike tariffs or import bans, this setting allows evasion to be observed in customs data.

An example illustrates the definition. In 2019, Russia imported 44 Ferrari cars, all recorded with Italy as the country of origin. In 2023, after exports of such cars to Russia were prohibited, six Ferrari cars still entered Russia, again with Italy listed as the country of origin.⁷ Under our definition, these transactions are classified as evasion.

Russian customs data separately report the country of origin and the country of dispatch. This allows us to identify transshipment routes and to distinguish between different forms of indirect trade. In the Ferrari example, the country of dispatch shifts from Italy before the bans to third countries after the bans, while the reported country of origin remains Italy. More generally, we use the country of dispatch to trace intermediary countries.

3 Stylized Facts about Evasion

Using our newly assembled data, we document five stylized facts about the magnitude and patterns of evasion of EU export bans on Russia. Together, these facts characterize both the extent of evasion and the channels through which it occurs, and they guide the structure of the quantitative model developed in the next section.

Fact 1: Banned EU exports continue to reach Russia Figure 4 plots monthly Russian imports of EU-origin products that subsequently become subject to EU export bans, measured in values and quantities. Import values rise in the months between Russia’s invasion of Ukraine in late February 2022 and the introduction of the export bans in July 2022, before falling sharply once the bans enter into force. This increase prior to July 2022 is consistent with stockpiling in anticipation of tighter restrictions, which is plausible given that the first sanctions were announced shortly after the invasion.

Despite this decline, Russian imports of EU-origin goods do not fall to zero, as would be implied by fully enforced export bans. Instead, they stabilize at a positive level and remain broadly flat through the end of 2024. The persistence of imports is not driven solely by price adjustments: quantities display a similar pattern. Since official EU export statistics report zero exports of these products after the bans enter into force, the continued appearance of EU-origin goods in Russian customs records necessarily reflects transactions that circumvent the export bans (Figure 3). As a simple summary measure of evasion, we compute the ratio of average monthly imports after July 2022 to average monthly imports in 2019. This ratio equals 26.0% for import values and 12.4% for quantities. These estimates constitute a lower bound on evasion: they exclude partially banned products, for which evasion may be easier because very similar non-banned variants exist, and they

⁷ Product descriptions indicate that these were new Ferrari cars rather than used vehicles.

capture only transactions in which the EU origin remains visible in the data, thus not accounting for any origin relabeling.

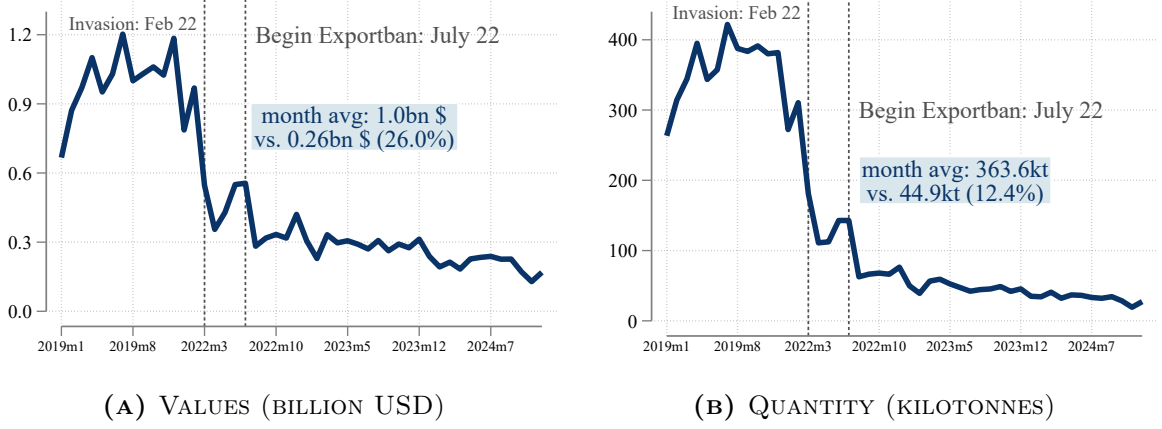


FIGURE 4. RUSSIAN IMPORTS OF EU-ORIGIN GOODS SUBJECT TO EXPORT BANS

Note: This figure plots the monthly value (billion USD) and quantity (kilotonnes) of Russian imports from the EU of products that later became subject to EU full export bans. Data are available for 2019 and for 2022–2024. Dashed vertical lines indicate Russia’s invasion of Ukraine in February 2022 and the introduction of the EU export bans in July 2022. After the export bans take effect, any remaining imports of these export-banned EU-origin products reflect evasion of the bans. We exclude products in HS2 88 (aircraft and parts) as they are extreme outliers.

Fact 2: Evasion occurs through distinct transshipment modes We use information on the country of dispatch to distinguish between different transshipment modes in the data. Figure 5a plots the share of Russian imports of EU-origin, export-banned products by dispatch country over time. Prior to the introduction of the export bans, the average monthly share of EU-origin shipments dispatched from EU countries is 87.4%, reflecting standard export patterns in which goods are shipped directly from the EU or routed through other EU countries before reaching Russia. Immediately after the export bans enter into force in July 2022, most evading shipments continue to be dispatched from EU countries. In this initial phase, evasion therefore predominantly occurs through transactions in which goods retain both an EU origin and an EU dispatch country, consistent with in-transit transshipment, where goods are redirected to Russia during shipping without a change in the exporter of record. Over time, however, the composition of transshipment modes shifts toward re-exporting. Starting after the export bans enter into force, the share of EU-origin imports dispatched from non-EU countries rises steadily to 60.6% by the end of 2024, with the steepest increase occurring between July 2022 and February 2023, when this share more than quadruples from 12.1% to about 50.5%.

Three patterns emerge from the evolution of transshipment modes. First, in-transit transshipment is particularly prevalent immediately after the export bans enter into force, when existing shipping arrangements continue to be used. Second, re-exporting

expands more gradually but steadily over time, eventually becoming the dominant mode of evasion, consistent with the presence of fixed costs involved in building up this mode of transshipment. Third, while re-exporting continues to grow after early 2023, its expansion slows markedly. This timing coincides with a period in which the EU and its partners intensified efforts to curb transshipment starting in March 2023, suggesting that stricter enforcement disproportionately affects the growth of re-exporting.

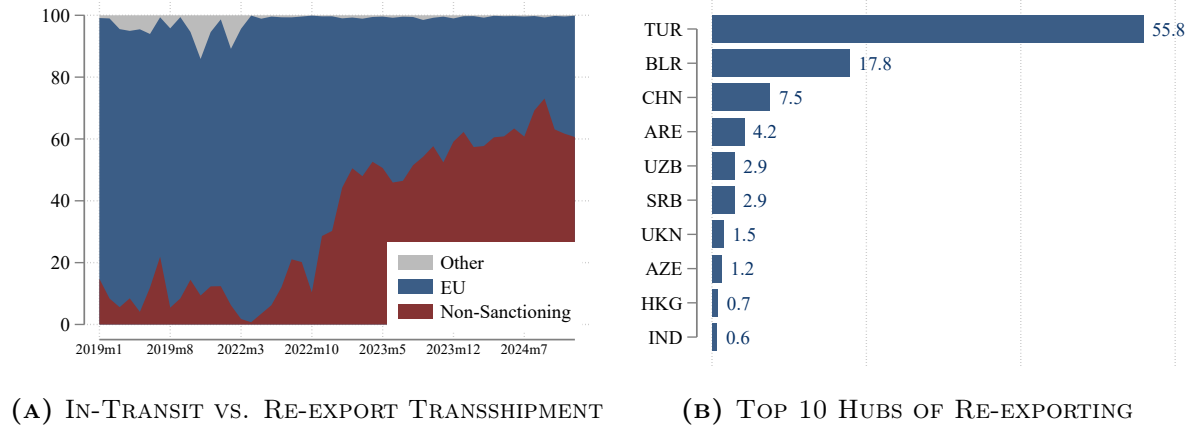


FIGURE 5. TRANSSHIPMENT MODES AND RE-EXPORTING HUBS

Note: Panel (A) shows the share of Russian imports of EU-origin, export-banned products by dispatch country over time, distinguishing in-transit transshipment (EU dispatch) from re-exporting transshipment (non-sanctioning dispatch). Panel (B) reports the top ten re-exporting hubs by their share of total re-exported trade. Hub names are reported as ISO-3 country codes; UKN denotes unknown.

Fact 3: Re-exporting is highly concentrated in a few hubs Re-exporting of EU-origin, export-banned products is highly concentrated geographically. Figure 5b shows the distribution of re-exported trade across intermediary countries. Three hubs—Türkiye, Belarus, and China—account for the majority of re-exporting activity. Türkiye alone accounts for more than half of total re-exported trade, while Belarus (17.8%) and China (7.5%) contribute substantial additional shares. At the same time, re-exporting is not confined to these three countries. An additional 18.9% of re-exported trade is routed through a broader set of intermediary hubs. Within the set of remaining intermediary hubs, re-exporting is itself concentrated: the United Arab Emirates, Uzbekistan, and Serbia together account for around 60% of re-exported trade routed through these hubs. The remaining share is distributed more evenly across a set of smaller intermediaries, including Azerbaijan, Hong Kong, India, Kyrgyzstan, Turkmenistan, Georgia, and Kazakhstan. Evasion through re-exporting therefore relies heavily on a small number of key hubs, while also exploiting a wider network of alternative routes.

Fact 4: Evasion varies widely across products Figure 6 illustrates substantial heterogeneity in evasion across products affected by the export bans. Panels (A) and (B)

plot the cumulative distribution of product-level evasion across the 688 fully export-banned HS6 products (by July 2022), where evasion is measured as the ratio of average monthly EU-origin imports in the post-bans period (August 2022–December 2024) to the average monthly imports in 2019. Values close to one reflect post-ban imports near their pre-war levels, values above one reflect imports exceeding their 2019 average, and values close to zero reflect a collapse in imports following the bans. Panel (A) reports the distribution using import values, while Panel (B) uses quantities. At the lower end of the distribution, 9.7% of products are no longer imported following the introduction of export bans, while a further 24.5% exhibit evasion rates below 20%, indicating that imports of these products decline to only a small fraction of their pre-war levels. At the upper end, 37.3% of products have evasion rates of at least 80%, including 29% with evasion rates above 100%, for which post-ban imports exceed their 2019 averages. A similar picture emerges for quantities.

We next relate this heterogeneity to pre-war product characteristics measured in 2019. Panels (C)–(F) of Figure 6 show binned scatter plots of product-level evasion against a set of characteristics that proxy for the ease with which shipments can be concealed within normal trade flows and for the availability of substitutes outside the EU. These characteristics include measures of market concentration on the supplier and buyer sides, indicators of prior third-country routing, and physical attributes related to bulkiness, proxied by product weight.

Evasion is higher for products characterized by less concentrated supplier markets, indicating the presence of a larger set of alternative suppliers, and for products purchased by a less concentrated set of Russian buyers which indicates that many trade relationships are already established. This pattern is consistent with evasion being easier for products that circulate in larger volumes through these existing networks, as such flows both provide cover within legitimate trade and benefit from firms’ existing know-how in handling the products, including familiarity with documentation, packaging, and customs procedures.

Evasion is also higher for products that were already routed through third countries prior to the bans, even at low levels, suggesting that existing third-country routing facilitates rerouting. This pattern is consistent with the presence of sunk costs in establishing transshipment routes: even limited pre-war use of intermediaries appears to lower the barriers to expanding indirect trade after the bans. In addition, evasion is systematically higher for lighter products, which are cheaper to divert and easier to conceal within normal trade flows. By contrast, unit values do not exhibit a systematic relationship with evasion, nor does the typical bundling of multiple HS6 products in shipments, which serves as a proxy for the ease of concealing goods within mixed consignments (Appendix Figure B.1).

Fact 5: The most heavily evaded products are hard to substitute A distinct pattern emerges for products in the extreme right tail of the evasion distribution, namely

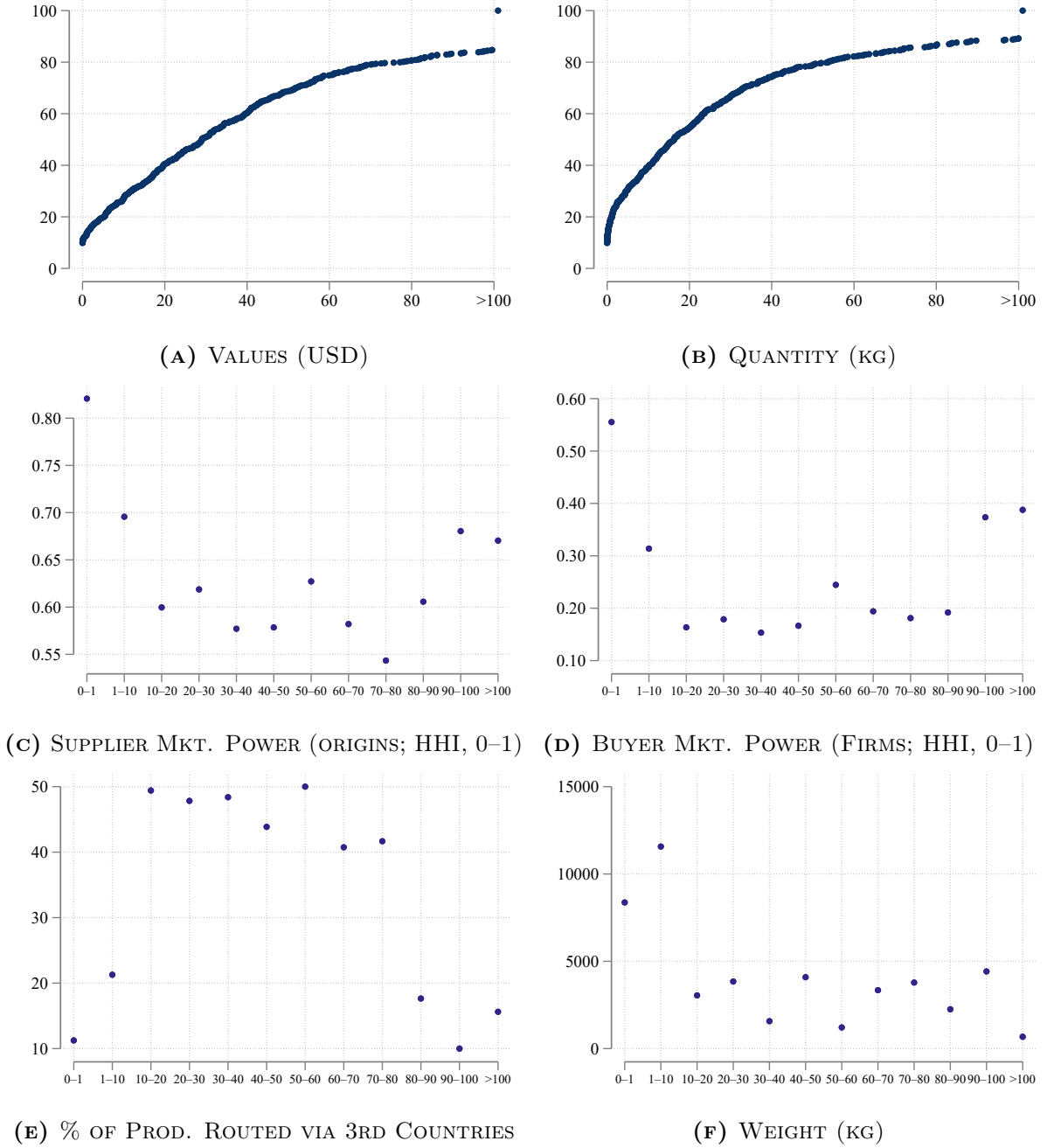


FIGURE 6. HETEROGENEITY IN EVASION ACROSS PRODUCTS

Note: Panels (A) and (B) plot the cumulative distribution of evasion across the 668 HS6 products subject to EU export bans introduced as of July 2022. Evasion is measured as the ratio of average monthly Russian imports of EU-origin products in the post-ban period (August 2022–December 2024) to the corresponding average in 2019; values above 100% indicate post-ban imports that exceed their 2019 levels. Panel (A) constructs the cumulative distribution of evasion using import values (USD), while Panel (B) uses quantities (kg). Panels (C)–(F) relate product-level evasion to pre-war product characteristics measured in 2019. In Panel (C), supplier market power is measured using the Herfindahl–Hirschman Index (HHI), computed across the product’s countries of origin. In Panel (D) buyer market power is measured using the Herfindahl–Hirschman Index (HHI), computed across the product’s importing firms. Each point reports the unweighted mean of the characteristic shown on the y-axis within a bin of the evasion measure, except for weight, which is reported as the median within each bin.

those for which post-ban imports exceed their 2019 levels. This group is quantitatively important: 29.0% of products in value terms and 18.3% in quantity terms fall into this category. Relative to other export-banned products, these top-evaded goods are characterized by higher concentration in the supplier market, as measured by an HHI computed across countries with the EU treated as a single supplier, and by a more concentrated set of Russian buyers, consistent with limited scope for substitution. At the same time, they exhibit moderate levels of pre-war third-country routing: on average, 25.6% of these products had at least one shipment in 2019 that was routed from the EU through a non-coalition intermediary before reaching Russia, a share similar to that of products with evasion rates below 20%. By contrast, products with high (though not above-2019 import-level) evasion rates exhibit substantially higher pre-war routing shares. For products with evasion rates between 70–90%, average routing shares range from 48.3–50.0%, before declining slightly to 46.4% for products with evasion rates between 90–100%.

Taken together, these patterns indicate that the most heavily evaded products combine high market concentration on the supplier and buyer sides with few pre-existing transshipment channels. For this subset of goods, continued access appears particularly important: Russia sustains imports despite the higher cost of establishing new routes. Evasion thus reflects the difficulty of substitution as well as the ease of concealment.

These facts shape the quantitative framework developed below. We work at the HS6 level so that Russia’s dependence on coalition suppliers, the availability of alternative suppliers, and pre-war third-country routing differ across products. We distinguish military-relevant, other critical, and remaining goods to allow substitution to vary across broad uses. Finally, we allow intermediary countries to differ in their ability to facilitate banned trade and to earn from doing so, consistent with the concentration of re-exporting in a small set of hubs.

Coalition group. (MOVE UP BEFORE THEORY SECTION? REDUCE?) In the counterfactual exercises below, we need to specify which countries sanctioned (banned) which products. These countries will form the “coalition.”

Following [Li et al. \(2024\)](#), we identify 38 U.S.-aligned countries exempt from U.S. licensing requirements for exports to Russia, as listed in Supplement No. 3 of Part 746 of the Export Administration Regulations (EAR), which recognizes countries implementing “substantially similar export controls.” Beyond the EU-27, several countries align with EU sanctions to varying degrees, through selective adoption, voluntary compliance, or formal implementation as part of EU accession. A core group—including Albania, Bosnia and Herzegovina, Iceland, Liechtenstein, Montenegro, North Macedonia, Norway, and Ukraine (IN THE MODEL I ALWAYS SET UKRAINE TO FULL EMBARGO?)—consistently

follows EU sanction packages.⁸ For these countries, we assume the EU sanction regime applies identically. For the remaining countries (see part III of Table 1), we rely on product-level sanction information from the Global Trade Alert (GTA) database. We retain sanctions at the 6-digit product level, applied to either all firms or individual firms, classified as discriminatory, and with an implementation date before December 2024. In the absence of explicit entry-into-force dates, we define sanctions using a two-step approach: (i) product–country observations are explicitly listed as sanctioned in GTA, and (ii) product–country–time observations for which exports to Russia fall below 10% of their pre-shock average.⁹ The EU’s sanction regime broadly encompasses the products targeted by these coalition countries (Table 2), justifying their designation as such. Nevertheless, with roughly 3,000 sanctioned products, the EU’s coverage extends beyond that of its coalition partners. For the remaining non-EU coalition countries, GTA provides product-level sanction flags but does not distinguish between partial and full bans across 6-digit codes. We therefore refrain from using these countries in analyses that require a precise full-versus-partial classification, such as in the triple-difference event-study design of Section 4.

TABLE 1. SANCTIONING COALITION COUNTRIES

EU-27			
AUT — Austria	BEL — Belgium	BGR — Bulgaria	CYP — Cyprus
CZE — Czechia	DEU — Germany	DNK — Denmark	ESP — Spain
EST — Estonia	FIN — Finland	FRA — France	GRC — Greece
HRV — Croatia	HUN — Hungary	IRL — Ireland	ITA — Italy
LTU — Lithuania	LUX — Luxembourg	LVA — Latvia	MLT — Malta
NLD — Netherlands	POL — Poland	PRT — Portugal	ROU — Romania
SVK — Slovakia	SVN — Slovenia	SWE — Sweden	—
Mirrors EU sanctions			
ALB — Albania	BIH — Bosnia & Herzegovina	ISL — Iceland	LIE — Liechtenstein
MKD — North Macedonia	MNE — Montenegro	NOR — Norway	UKR — Ukraine
Other sanctioning countries			
AUS — Australia	CAN — Canada	CHE — Switzerland	GBR — UK
JPN — Japan	KOR — South Korea	NZL — New Zealand	SGP — Singapore
TWN — Taiwan	USA — United States	—	—

Note: The table distinguishes three groups of sanctioning countries. The first group comprises the EU-27 member states. The second group includes non-EU countries that closely mirror the EU sanctions regime and for which the same sanctions are applied consistently. The third group consists of non-EU coalition countries that have imposed their own sanctions regimes on Russia, which differ to varying degrees from the EU regime.

The EU has also extended its sanction regime to Belarus in response to the country’s involvement in Russia’s military aggression against Ukraine. This is important for our quantification, because we will block transshipment via Belarus in the model for the products where it was also hit by sanctions. Sanction data are drawn from TARIC, applying the same inheritance rules as for EU-Russia sanctions. By July 2022, 1,195

⁸ [Council alignment statement, December 8, 2025](#)

⁹ For product–origin pairs absent from the pre-period, we proxy the EIF date with the implementation date.

TABLE 2. COVERAGE OF EU-RUSSIA SANCTIONS BY NON-EU COALITION COUNTRIES

Country	Share of products covered by EU’s regime [%]	Total products sanctioned	Overlap with EU
Singapore	100.0	15	15
Switzerland	99.8	432	431
Canada	98.6	148	146
South Korea	97.7	386	377
United Kingdom	96.9	294	285
USA	88.9	416	370
Japan	74.2	458	340
Australia	70.0	30	21
New Zealand	42.9	7	3

Note: This table reports, for each country in the non-EU coalition group, the extent to which its sanctions regime with respect to Russia is covered by the EU-Russian sanction regime, measured as the share of sanctioned products that are also sanctioned by the EU vis-a-vis Russia. Columns (2) and (3) report the total number of products sanctioned by each country and the number of those products that are also sanctioned by the EU.

products were subject to export bans, rising to 1,299 products by December 2024 (see Figure A.1). The overlap with EU-Russia sanctioned products was initially limited: only 29% of products fully sanctioned against Russia by July 2022 were also fully sanctioned vis-à-vis Belarus. By December 2024, this share increased to approximately 38%.¹⁰ The sanctioned products for the remaining coalition countries are again taken from Global Trade Alert (GTA).

4 Estimating the Actual Trade-Cost Increase

SHORTEN AND MOVE UP – MUST COME BEFORE THE TRADE-COST CALIBRATION

To quantify the de facto increase in trade costs implied by the export bans, we bring the model’s gravity structure to the data. The object of interest is the change in bilateral trade costs for fully sanctioned EU products when exporting to Russia. We estimate this using a triple-difference (DDD) event-study design at the HS6–origin–month level for 2019 and 2022–2024. We restrict the sample to products that were either never sanctioned or fully banned by July 2022, when the first major wave of export bans became binding. This avoids complications due to staggered treatments and ensures a clean comparison between treatment and control observations. By excluding partly sanctioned products, we can cleanly identify actual evasion, rather than potentially misinterpreting trade in non-sanctioned product subgroups as such.

The estimating equation follows directly from equation (11), where export bans enter as an increase in iceberg trade costs for banned products. Combining $X_{e,RUS}^k = \pi_{e,RUS}^k X_{RUS}^k$ with equation (11) and taking logs yields,

¹⁰ For EU–Belarus sanctions, the implementation date rather than the entry-into-force date is used to define sanctioned products. Consequently, these overlap figures likely represent an upper bound, as wind-down periods and exemptions may delay the actual enforcement of the sanctions.

$$\log(X_{e,RUS}^k) = \underbrace{\log(T_e^k w_e^{-\theta^k})}_{\text{exporter-specific}} + \underbrace{\log(X_{RUS}^k (P_{RUS}^k)^{\theta^k})}_{\text{importer-specific}} - \underbrace{\theta^k \log(\tau_{e,RUS}^k)}_{\text{bilateral trade cost}}, \quad (1)$$

where the first and second elements on the right-hand side are exporter- and importer-specific, respectively. The only bilateral component left is the trade cost, whose net increase is precisely what we want to measure. In other words, the gravity theory guides us in setting up exporter- and importer-specific controls to isolate trade-cost changes, such that the treatment coefficients below identify changes in $-\theta^k \log(\tau_{e,RUS}^k)$. To then translate the increase in $\tau_{e,RUS}^k$ into aggregate effects, we will use the full GE framework in Section 6.¹¹ For ease of notation, we drop the *RUS* subscript for the remainder of this section.

Guided by equation (1), we set up an empirical counterpart that respects this structure and accounts for the temporal dimension in the data. Our empirical design compares (i) fully sanctioned versus never-sanctioned products, (ii) EU (and mirroring) versus non-coalition origins, and (iii) pre- versus post-ban periods. Identification therefore comes from the relative decline in EU exports of fully sanctioned products to Russia after the bans, net of changes for non-sanctioning countries and for products never subject to restrictions. Specifically, the regression specification is given by

$$X_{et}^k = \exp \left[\sum_{m=\text{Feb 2019}}^{\text{Dec 2024}} \beta_m S^k \cdot S_e \cdot \mathbf{1}\{\text{event}_t = m\} + \mu_e^k + \mu_{et} + \mu_t^k \right] + \epsilon_{et}^k, \quad (2)$$

where X_{et}^k denotes Russian imports from origin e of HS6 product k in month t , $S^k = 1$ for fully sanctioned products, and $S_e = 1$ for exporters that are part of the sanctioning coalition. The coefficient path $\{\beta_m\}$ traces the evolution of sanctioned EU export flows relative to the three comparison groups described above. As is standard in the trade literature, we follow Santos Silva and Tenreyro (2006) and estimate equation (2) in its multiplicative form using Poisson Pseudo-Maximum Likelihood (PPML). This avoids biased and inconsistent estimates in the presence of heteroskedasticity due to the log-linearization necessary for OLS estimation and can accommodate zero flows. This feature is particularly important in our context, as export bans precisely aim to lower the respective flows all the way to zero, and observations corresponding to fully successful bans would be dropped if a log-linear counterpart of (2) were estimated instead.

The specification includes three sets of high-dimensional fixed effects with a clear structural interpretation relating to a multi-period stochastic counterpart of the theoretical gravity equation (1). Product-origin fixed effects, μ_e^k , absorb time-invariant technological

¹¹ Note that we measure the average change in $\theta^k \log(\tau_{e,RUS}^k)$, which we will translate into $\tau_{e,RUS}^k$ using the trade elasticities, θ^k , described in Section 5.3.

differences, T_e^k , determining comparative advantage across and absolute advantage within products, as well as time-invariant geographical and non-sanction-related trade policy determinants of exporter-product-specific trade costs with Russia. Origin-month fixed effects μ_{et} capture the exporter’s competitiveness in terms of its wage, w_{et} , as well as origin-wide trade cost shocks with Russia that affect all products, such as reputational risk or logistical and financial constraints (including SWIFT exclusion).¹² Product-month fixed effects, μ_t^k , absorb time-varying Russian demand at the product level, $X_{RUS,t}^k$, as well as the product-specific competitive environment in the Russian market, as captured by the price index $P_{RUS,t}^k$, or equivalently by the so-called inward multilateral resistance term in a gravity framework, which formalizes how accessible the Russian market for the respective product is overall for suppliers from all origins.

Figure 7 reports the estimated event-study coefficients, normalizing January 2019 to zero. We find no evidence of differential pre-trends. Exports begin to decline when the war starts, but the sharp and persistent drop occurs in July 2022, the first month in which the initial wave of export bans is fully in force. This pattern is consistent with the long transition periods built into the sanctions regime: although the bans were announced earlier, many only became legally binding in mid-2022.¹³

A useful benchmark is the case of perfect enforcement. Under a gravity structure like ours, a fully enforced export ban corresponds to an infinite bilateral trade cost, which would imply that sanctioned EU exports fall to zero in Russian customs data. In that case, the trade-cost increase would be unidentified. The large—but far from complete—decline observed in the data therefore pins down the finite de facto increase in trade costs implied by the residual flows. Our triple-difference design exploits precisely this remaining variation.

Our baseline estimate implies that the export bans reduced EU exports of fully sanctioned products to Russia by roughly 80 percent, relative to the control flows. This is a substantial decline, but it falls short of the complete cessation implied by perfect enforcement, consistent with the extensive evasion documented above. In the calibrated model, this estimate implies that the bans raise median effective trade costs from the EU to Russia by about 69%, far below the prohibitive increase under full enforcement.

¹² Technically, to control for the impact of w_{et} on trade, we would need origin-product-month fixed effects, because w_{et} enters the gravity equation with a product-specific elasticity, θ^k . However, this would absorb all the variation in X_{et}^k .

¹³ Figure B.2 in the Appendix shows the results from the same specification (2) as Figure 7, but with export quantities rather than export values as the dependent variable. The pattern is very similar, with a somewhat stronger average effect size of -85% .

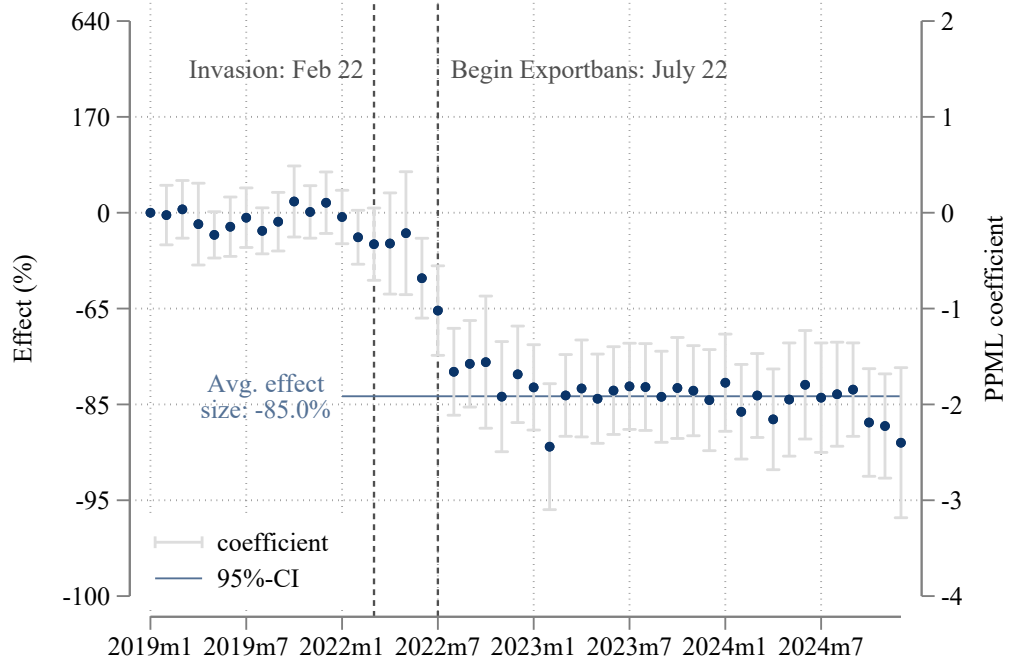


FIGURE 7. BASELINE ESTIMATION RESULTS: EVENT-STUDY

Note: The figure shows the estimated $\hat{\beta}_m$ coefficients of the event-study specification (2). The coefficient for the first month (January 2019) is normalized to zero. The raw coefficients are shown on the right y-axis. The left y-axis reports the corresponding partial-equilibrium trade-flow changes, $(e^{\hat{\beta}_m} - 1) \times 100\%$.

5 A Quantitative Model of International Trade and Transshipment

We develop a general-equilibrium model of international trade with endogenous transshipment. The framework combines three elements. First, a multi-product [Eaton and Kortum \(2002\)](#) (henceforth, EK) structure captures substitution across source countries within narrowly defined products. Second, nested final demand distinguishes military-relevant goods, other critical goods, and all remaining goods, allowing substitution across these broad groups to be harder than substitution across products within a group. Third, following an export ban, coalition-origin goods can be rerouted through third-country intermediaries that differ in their comparative advantage in facilitating transshipment and may earn rents from doing so.

The model distinguishes legal trade diversion (i.e., substitution toward non-sanctioning producers) from evasion, where a banned product continues to reach Russia through an intermediary. We use the framework to quantify the effects of export bans on trade, real income, and access to military-relevant and critical goods, and to study how rerouting opportunities and hub rents affect sanctions design.

5.1 The Baseline Economy

The framework builds on the multi-product version of EK (see [Costinot et al., 2012](#); [French, 2016](#)). Countries trade intermediate varieties that are assembled into a finite set of non-tradable final products. Countries are indexed by $e, i \in \{1, 2, \dots, N\}$, where e denotes the exporter and i the importer. Products are indexed by $k \in \{1, 2, \dots, K\}$, while ω^k indexes the tradable intermediate varieties used to assemble product k .

We divide products into three mutually exclusive and exhaustive groups, $\mathcal{G} = \{M, C, O\}$, where M denotes military-relevant goods, C other critical goods, and O all remaining goods. Let $\mathcal{K}_g$ denote the set of products assigned to group g , and let $g(k)$ denote the group containing product k .

Labor is the only production input, and households supply it inelastically to competitive domestic producers. We first describe final demand and then turn to production, trade, and equilibrium.

5.1.1 Demand Side: Households

Country i is populated by L_i households, each of which inelastically supplies one unit of labor to domestic producers. For tractability, we represent their aggregate final absorption

through a representative household.¹⁴ The representative household has CES preferences over the three non-tradable group composites,

$$U_i = \left[\sum_{g \in \mathcal{G}} (\alpha_i^g)^{1/\nu} (Q_i^g)^{\frac{\nu-1}{\nu}} \right]^{\frac{\nu}{\nu-1}}, \quad \nu > 0, \quad (3)$$

where $\alpha_i^g > 0$ is a group-specific demand weight, normalized such that $\sum_g \alpha_i^g = 1$, and ν is the elasticity of substitution across the three broad groups (M , C , and O).

Each group composite is assembled from the products assigned to that group, $\mathcal{K}_g$,

$$Q_i^g = \left[\sum_{k \in \mathcal{K}_g} (q_i^k)^{\frac{\sigma_g-1}{\sigma_g}} \right]^{\frac{\sigma_g}{\sigma_g-1}}, \quad \sigma_g > 0, \quad (4)$$

where σ_g governs substitution across products within group g . Finally, every product k is composed of a continuum of k -specific tradable varieties, with the measure of varieties normalized to one,

$$q_i^k = \left[\int_0^1 (q_i^k(\omega^k))^{\frac{\eta^k-1}{\eta^k}} d\omega^k \right]^{\frac{\eta^k}{\eta^k-1}}.$$

We assume $\eta^k > \sigma_{g(k)} \geq \nu > 0$, with the usual limiting expressions when an elasticity equals one. Thus, varieties within a specific product are closer substitutes than different products, while products within the same broad group can be closer substitutes than products with fundamentally different uses. When $\sigma_g = \nu = \sigma$ for all groups, the two upper tiers collapse to a single CES aggregator over products, with each product inheriting the demand weight of its group.

The representative household maximizes utility subject to the budget constraint $E_i \geq \sum_{k=1}^K \int_0^1 p_i^k(\omega^k) q_i^k(\omega^k) d\omega^k$, where E_i is aggregate income and $p_i^k(\omega^k)$ is the minimum delivered price of variety ω^k across all potential source countries, as explained in the next subsection. Letting P_i^k , P_i^g , and P_i denote the product-, group-, and aggregate price indices in country i , respectively, the relevant elasticity at each tier is η^k within product k , σ_g across products in group g , and ν across the three groups. Expenditure minimization

¹⁴ This is only an expositional convention. Equivalently, competitive assemblers could purchase the tradable varieties, combine them first into product k and then into the three group composites M , C , and O , and sell those composites to the representative household. With constant returns and zero profits, the assemblers' unit costs are exactly the price indices derived below, so the demand and equilibrium conditions are unchanged.

yields

$$P_i^k = \left[\int_0^1 (p_i^k(\omega^k))^{1-\eta^k} d\omega^k \right]^{1/(1-\eta^k)}, \quad (5)$$

$$P_i^g = \left[\sum_{k \in \mathcal{K}_g} (P_i^k)^{1-\sigma_g} \right]^{1/(1-\sigma_g)}, \quad (6)$$

$$P_i = \left[\sum_{g \in \mathcal{G}} \alpha_i^g (P_i^g)^{1-\nu} \right]^{1/(1-\nu)}. \quad (7)$$

Writing group-level expenditure as $X_i^g = P_i^g Q_i^g$ and product-level expenditure as $X_i^k = P_i^k q_i^k$, with $X_i = \sum_{g \in \mathcal{G}} X_i^g = \sum_{k=1}^K X_i^k$, the expenditure shares are

$$\frac{X_i^g}{X_i} = \alpha_i^g \left(\frac{P_i^g}{P_i} \right)^{1-\nu}, \quad (8)$$

$$\frac{X_i^k}{X_i} = \left(\frac{P_i^k}{P_i^g} \right)^{1-\sigma_g}, \quad k \in \mathcal{K}_g. \quad (9)$$

Combining the two expressions,

$$\frac{X_i^k}{X_i} = \alpha_i^{g(k)} \left(\frac{P_i^{g(k)}}{P_i} \right)^{1-\nu} \left(\frac{P_i^k}{P_i^{g(k)}} \right)^{1-\sigma_{g(k)}}. \quad (10)$$

The upper nest governs substitution across the three broad product groups. For a given increase in a group's relative price, a lower ν causes its expenditure share to fall less. The model therefore allows Russia to protect expenditure on strategically important products when they become more costly, rather than treating substitution between, for example, military inputs and ordinary consumer goods as equivalent to substitution between products with similar uses.¹⁵

Next, we outline how the intermediate varieties are produced and shipped.

5.1.2 Supply Side: Production of Tradable Varieties

Intermediate varieties are produced and shipped by perfectly competitive firms using labor as the only input in production.

Trade costs. Throughout the paper, we index the exporter by e and the importer by i . A subscript ei can be read as “from country e to i .” International trade is costly:

¹⁵ The military-relevant composite is a model-based measure of real access to that product group. It should not be interpreted as a direct measure of military capability or as a military production function.

intermediate inputs are traded subject to standard iceberg trade costs. Firms in e need to ship $\tau_{ei}^k \geq 1$ units of a variety ω^k for one unit to arrive in i . We normalize $\tau_{ii}^k = 1$ and assume that the so-called triangle inequality holds, such that $\tau_{ei}^k \leq \tau_{eh}^k \tau_{hi}^k \forall (e, h, i, k)$. This no-arbitrage condition will be central in the transshipment analysis, as detailed in Section 5.2.

Production technology. Labor is perfectly mobile across industries but immobile across countries, which yields a common wage across industries. Firms operate in perfect competition with constant returns to scale, and let $\varphi_e^k(\omega^k)$ denote the labor productivity in country e to produce variety ω^k . Hence, the cost of producing one unit of ω^k and shipping it to i is given by the marginal cost, $c_{ei}^k(\omega^k) := (w_e \tau_{ei}^k) / \varphi_e^k(\omega^k)$, where w_e is the wage rate in country e . Note that $c_{ei}^k(\omega^k)$ is also the price offered by e to i under perfect competition.

Following the literature, $\varphi_e^k(\omega^k)$ is independently drawn from a Fréchet distribution, $F_e^k(\varphi) = \exp(-T_e^k \varphi^{-\theta^k})$, for $\varphi > 0$. T_e^k measures the overall technology level in a country e to produce the varieties for product k , while θ^k governs the forces of comparative advantage across varieties of a product, as it is inversely related to the dispersion of productivities.

5.1.3 Trade Flows and Equilibrium

Trade. Consumers pick the lowest-cost source of a variety across the world. This, coupled with the Fréchet distribution, yields the familiar gravity equation of trade flows (i.e., the expenditure share in country i on varieties from e and product k)¹⁶

$$\pi_{ei}^k := \frac{X_{ei}^k}{X_i^k} = \frac{T_e^k (w_e \tau_{ei}^k)^{-\theta^k}}{\sum_{l=1}^N T_l^k (w_l \tau_{li}^k)^{-\theta^k}}, \quad (11)$$

where X_{ei}^k is country i 's expenditure on product- k varieties sourced from e , and $X_i^k := \sum_{e=1}^N X_{ei}^k$ is country i 's total expenditure on product k . We refer to π_{ei}^k as the “trade share.” Intuitively, trade flows are higher, the more productive e is on average (higher T_e^k), and the lower its production and transportation costs (lower $w_e \tau_{ei}^k$) relative to the other countries. We refer to θ^k as the “trade elasticity,” because a higher θ^k implies that trade flows are more elastic with respect to changes in trade costs. After a product-specific normalization of technology, the denominator in equation (11) gives the product-level

¹⁶ This follows from standard derivations (see Costinot et al., 2012; Eaton and Kortum, 2002).

price index,¹⁷

$$P_i^k = \left[\sum_{e=1}^N T_e^k (w_e \tau_{ei}^k)^{-\theta^k} \right]^{-1/\theta^k}. \quad (12)$$

Note that the nested final-demand structure affects how total expenditure is allocated across and within product groups, but it does not change the sourcing problem conditional on product k . The gravity equation and the product-level EK price index therefore retain their standard form with and without upper-tier demand nests.

Baseline equilibrium. Because firms are perfectly competitive and labor is the only production input, labor-market clearing requires producer income to equal total expenditure on the country's goods,

$$w_i L_i = \sum_{j=1}^N \sum_{k=1}^K \pi_{ij}^k X_j^k. \quad (13)$$

Labor is also the only source of household income in the baseline, so $X_i = w_i L_i$. Together with the nested expenditure shares, these conditions determine the baseline wage vector up to a numéraire. They also imply the standard balanced-trade condition, i.e., after domestic expenditure is netted out, the value of each country's total imports equals the value of its total exports.

5.2 Export Bans and Transshipment

Having characterized the baseline trade economy, we now introduce export bans and transshipment. We proceed in two steps. First, we use the triangle inequality (TI) to recover the minimum post-sanction rerouting costs implied by the calibrated international trade network. Second, we allow intermediary countries to differ in their comparative advantage in facilitating banned trade and permit the selected intermediary to charge a markup. The first step isolates the rerouting implied by the pre-war trade network alone, while the second helps the model account for the observed distribution of transshipment hubs and gives intermediaries a private return from facilitating evasion. For notational convenience, let R denote Russia throughout this subsection.

¹⁷ Before this normalization, the exact price index is

$$P_i^k = A_k \left[\sum_{e=1}^N T_e^k (w_e \tau_{ei}^k)^{-\theta^k} \right]^{-1/\theta^k}, \quad A_k := \Gamma \left(\frac{\theta^k + 1 - \eta^k}{\theta^k} \right)^{1/(1-\eta^k)}.$$

Because A_k (consisting of the Gamma function, $\Gamma(\cdot)$) is common to all origins and destinations within product k , we can define $\tilde{T}_e^k := A_k^{-\theta^k} T_e^k$. This rescaling leaves the trade shares in equation (11) unchanged and absorbs A_k into the technology term. Relabeling $\tilde{T}_e^k$ as T_e^k gives equation (12) with equality. We assume $\theta^k > \eta^k - 1$, which ensures that the product-level price index is finite.

5.2.1 The Triangle Inequality and Benchmark Rerouting

To measure transshipment opportunities, we rely on the triangle inequality of trade costs,

$$\tau_{ei}^k \leq \tau_{eh}^k \tau_{hi}^k \quad \forall (e, h, i, k), \quad (14)$$

building on recent insights from Foellmi et al. (2024). In words, the TI is a no-arbitrage condition: the trade cost assigned to supplying product k from country e to country i cannot exceed the cost of supplying the same product through an intermediary h . Otherwise, firms would use the cheaper indirect route, and the effective bilateral trade cost would become $\tau_{eh}^k \tau_{hi}^k$.¹⁸

The TI becomes particularly important after a shock. Starting from the TI-consistent baseline matrix τ , let $\tilde{\tau}'$ denote the trade-cost matrix immediately after the direct policy shock, before rerouting is taken into account. For a coalition origin e that bans exports of product k to Russia, $\tilde{\tau}_{eR}^{k'} \rightarrow \infty$, while bilateral costs that are not directly affected by the policy remain at their baseline values.

The direct-shock matrix $\tilde{\tau}'$ need not satisfy the TI. If, for some potential intermediary h , $\tilde{\tau}_{eR}^{k'} > \tilde{\tau}_{eh}^{k'} \tilde{\tau}_{hR}^{k'}$, the direct post-sanction cost cannot be the effective trade cost, and we replace it by the cheaper indirect cost. Denote the resulting TI-consistent post-shock matrix by $\bar{\tau}_{ei}^{k'}$. By construction, no remaining indirect route is cheaper than the corresponding bilateral cost recorded in $\tilde{\tau}'$. This matrix constitutes our benchmark rerouting technology: it gives the minimum post-sanction trade costs implied by the calibrated international trade network, before introducing any sanctions-specific hub advantage or intermediation rent.¹⁹ In the benchmark without the additional hub technology introduced below, the relative trade-cost change used in the counterfactual exercise would therefore be $\bar{\tau}_{ei}^{k'} / \tau_{ei}^k$.

This logic highlights why calibrated trade-cost *levels* are required. Knowing only that the direct coalition–Russia cost becomes prohibitive does not reveal whether Türkiye, China, the United Arab Emirates, or another country is the cheapest intermediary. Ranking these alternatives and measuring the trade-cost increase from rerouting requires the levels

¹⁸ This condition can be microfounded using link-level trade costs (Foellmi et al., 2024). In that interpretation, the bilateral cost τ_{ei}^k is the cost of the cheapest available path from e to i , rather than necessarily the cost of a direct shipment. The TI therefore does not imply that all observed trade moves directly between the origin and destination. It implies only that the reduced-form bilateral trade costs used in the gravity equation already incorporate any cost-saving indirect routes that firms use in equilibrium.

¹⁹ In general, the updating is repeated because the cheapest route to an intermediary may itself use another country. Iterating until no TI is violated incorporates one-hub and multi-leg paths into a TI-consistent post-shock bilateral cost matrix, and Foellmi et al. (2024) show that the procedure converges to a unique solution after a finite number of iterations. In our application, the policy raises only banned coalition-origin links to Russia, while the links of eligible non-coalition intermediaries remain unchanged. Starting from a TI-consistent baseline, any route involving several eligible intermediaries can then be weakly shortened to a one-hub route. The bilateral legs in $\tilde{\tau}'$ may nevertheless themselves summarize multi-link transportation paths.

of the two relevant bilateral costs, τ_{eh}^k and τ_{hR}^k , for every candidate hub. This is why the calibration below recovers a TI-consistent matrix of baseline trade-cost levels rather than relying only on relative changes.

The TI benchmark also makes the distinction between trade diversion and evasion precise. If Russia responds to an export ban by buying product k from a producer in a non-sanctioning country, the adjustment is standard trade diversion. If the product remains of coalition origin but reaches Russia through an intermediary after the direct export has been prohibited, the transaction circumvents the ban and constitutes evasion.

This benchmark abstracts from two features that are central to the present application. First, hub choice is determined entirely by the calibrated pre-war trade network and therefore need not reproduce the observed relative importance of intermediary countries after the sanctions. Second, intermediation is perfectly competitive, so the country through which the product is rerouted earns no rent from facilitating the transaction. The latter implication is unattractive when studying the incentives of intermediary countries. If facilitating transshipment generates no private benefit, any positive expected penalty makes participation strictly unattractive. We therefore enrich the TI benchmark with a sanctions-specific hub advantage and an intermediation markup.

5.2.2 Hub Comparative Advantage and Rents

Consider a banned shipment of product k from coalition origin e to Russia. For every eligible intermediary h ,²⁰ define the adjusted route cost

$$m_{h,eR}^k = \max \left\{ \bar{\tau}_{eh}^{k'} \bar{\tau}_{hR}^{k'} \rho_h, 1 \right\}, \quad (15)$$

where $\rho_h > 0$ is a hub-specific comparative-advantage wedge. A lower ρ_h makes country h more attractive for rerouting banned goods, conditional on the TI-based benchmark costs. The lower bound in equation (15) ensures that the iceberg component of the route cost cannot fall below one.

We assume that ρ_h is specific to post-sanction intermediation. It captures differences in hub attractiveness that are not already reflected in the TI-based benchmark costs, including commercial and logistics networks, as well as documentation and concealment capacity. It does not modify the pre-war trade-cost matrix and does not apply to non-banned trade.

Let $m_{(1),eR}^k$ and $m_{(2),eR}^k$ denote the lowest and second-lowest adjusted route costs across eligible intermediaries. Potential hubs compete to provide the intermediation service. The country with the lowest adjusted cost is selected, while the next-best route limits the

²⁰ In the quantitative exercises, coalition countries cannot serve as hubs, and we exclude intermediary routes that are themselves blocked by the relevant export bans. Section 5.3 describes these restrictions in detail.

premium that it can charge without losing the transaction. This introduces a Bertrand-like pricing structure.²¹

The gross markup charged by the selected hub is

$$\kappa_{eR}^k(\beta) = 1 + \min \left\{ \beta \left(\frac{m_{(2),eR}^k}{m_{(1),eR}^k} - 1 \right), \frac{1}{\theta^k} \right\}, \quad \beta \in [0, 1]. \quad (16)$$

The first term inside the minimum captures route competition: β determines how much of its cost advantage the selected hub can extract. The second term imposes a parsimonious demand-based cap on the markup.²² Figure 8 schematically illustrates the triangle inequality and the resulting transshipment technology.

When $\beta = 1$, the route-competition component permits the selected hub to extract its full advantage over the second-best route (whenever this does not exceed the demand-based cap), whereas when $\beta = 0$, it earns no markup. Intermediate values allow the hub to capture only part of the available route advantage. Our benchmark quantification uses $\beta = 1$, but we show below that when recalibrating assuming $\beta = 0$, the main trade and real-income results are essentially unchanged, while hub rents disappear by construction.

The final importer-facing cost of the rerouted flow is

$$\tau_{eR}^{k'} = m_{(1),eR}^k \kappa_{eR}^k(\beta). \quad (17)$$

For bilateral product cells to which the hub technology does not apply, we set $\tau_{ei}^{k'} = \bar{\tau}_{ei}^{k'}$ and $\kappa_{ei}^k = 1$. In counterfactual exercises in which transshipment is ruled out entirely, the final trade cost of a banned coalition-origin flow to Russia remains prohibitive, $\tau_{eR}^{k'} \rightarrow \infty$.

The relative values of ρ_h primarily determine which countries are selected as intermediaries, hence the comparative-advantage interpretation, and we discipline them using the observed distribution of hub use. Their common scale governs the overall cost of rerouting. We calibrate it, taking into account the resulting route choices and markups, so that the model reproduces the trade-cost increase implied by the DiD estimates. Section 5.3 explains the calibration in detail.

²¹ This follows the Bertrand-pricing logic in Bernard et al. (2003), where the lowest-cost supplier serves the market, while the cost of the second-best supplier constrains the price it can charge.

²² The route-competition markup need not maximize the hub's revenue, because a higher markup reduces demand for the rerouted flow. To obtain a tractable cap, hold Russian expenditure on product k , wages, and competing origins' delivered costs fixed, and let the hub take the Russian product price index as given. From the gravity equation (11), multiplying its delivered cost by κ then implies that gross bilateral expenditure satisfies $X_{eR}^k(\kappa) \propto \kappa^{-\theta^k}$. The hub retains the fraction $1 - 1/\kappa$ of this expenditure. Maximizing $(1 - 1/\kappa)\kappa^{-\theta^k}$ with respect to κ yields $\kappa^* = 1 + 1/\theta^k$, so the proportional markup $\kappa - 1$ is capped at $1/\theta^k$. This is a tractable constant-elasticity partial-equilibrium benchmark rather than the exact solution to the full strategic pricing problem.

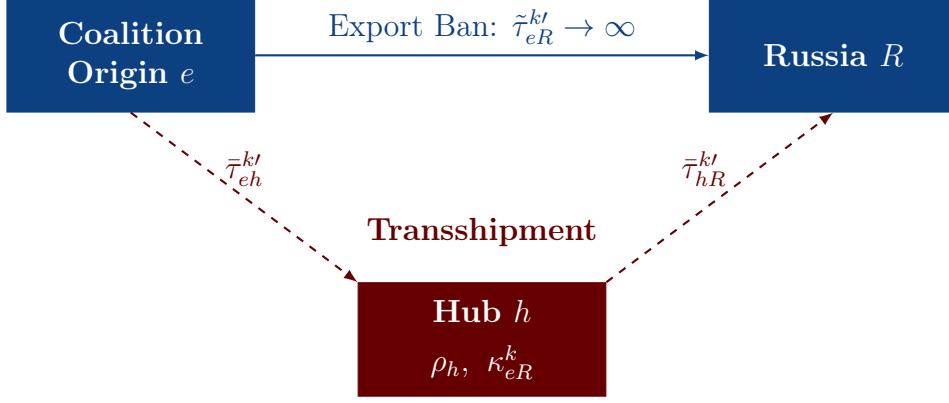


FIGURE 8. TRANSSHIPMENT WITH HUB COMPARATIVE ADVANTAGE AND MARKUPS

Notes. When the direct export of product k from coalition origin e to Russia is banned, the product can be rerouted through an intermediary h . The TI-consistent costs $\tilde{\tau}_{eh}^{k'}$ and $\tilde{\tau}_{hR}^{k'}$ summarize the two benchmark legs. Candidate hubs differ in the comparative-advantage wedge ρ_h , and the selected hub charges the markup κ_{eR}^k in equation (16).

Note that our approach matches the structure of the customs data. We observe the country of origin, the dispatch country, and the destination, which correspond to e , h , and R in the model. We do not model the full sequence of transport links between these countries (see, e.g., [Allen and Arkolakis, 2022](#); [Ganapati et al., 2024](#)), because we neither observe nor aim to explain every port or transport node along the route. The bilateral legs in Figure 8 are reduced-form trade costs and may themselves summarize more complicated transportation paths. We therefore use the dispatch country as the empirical counterpart to the model's selected intermediary.

The deterministic transshipment technology selects one hub for each origin-product cell, which provides a good approximation of the Russian customs data. Among the fully banned origin-product cells from EU origins and mirroring countries whose 2024 shipments reached Russia through non-coalition third countries, 53% use exactly one dispatch country. Conditional on using more than one hub, the median share of that routed value passing through the main dispatch country is 73%. We therefore interpret the model's selected hub as the dominant intermediary for an origin-product cell.

5.2.3 Solving the Model in Changes

The counterfactual equilibrium differs from the baseline in two respects. First, importers make their sourcing decisions using the final importer-facing costs τ' , which combine rerouting costs and, where applicable, hub markups. Second, the markup changes the international allocation of income. The importer pays the gross value of the rerouted shipment, the original exporter receives this value net of the markup, and the intermediary

receives the difference. We first describe this income accounting and then outline the full equilibrium system in relative changes.

Income accounting with hub rents. Let $X_{ei}^{k'}$ denote importer i 's gross expenditure on product k from origin e in the counterfactual equilibrium. For an untreated flow, $\kappa_{ei}^k = 1$, and the exporter receives the full amount. For a rerouted sanctioned flow, the original exporter receives $X_{eR}^{k'}/\kappa_{eR}^k$, while the selected hub receives $(1 - 1/\kappa_{eR}^k) X_{eR}^{k'}$. These two components sum to the importer's gross expenditure. We treat the markup as pure rent, so it reallocates income between the exporter and the hub rather than using additional resources. This differs from the role of ρ_h , which changes the non-rent component of the delivered trade cost.

Let $h^*(e, k)$ denote the top intermediary attaining $m_{(1),eR}^k$. Premium income accruing to hub country h is

$$R'_h = \sum_{e,k: h^*(e,k)=h} \left(1 - \frac{1}{\kappa_{eR}^k}\right) X_{eR}^{k'}. \quad (18)$$

We assume that this income is rebated to the representative household of the hub country. For intuition, note that the accounting is analogous to tariff revenue in a standard quantitative trade model such as [Caliendo and Parro \(2015\)](#). Writing the gross tariff factor as $\kappa = 1 + t$, importer expenditure is split into exporter income $X/(1 + t) = X/\kappa$ and tariff revenue $tX/(1 + t) = (1 - 1/\kappa)X$. Here, the same wedge accrues to the intermediary country rather than to the importing government.

Because labor is the only production input, the wage bill in exporter e equals its sales net of hub markups

$$w'_e L_e = \sum_{i=1}^N \sum_{k=1}^K \frac{X_{ei}^{k'}}{\kappa_{ei}^k}. \quad (19)$$

Total expenditure in country i is labor income plus any premium income earned as a hub,

$$X'_i = w'_i L_i + R'_i. \quad (20)$$

These equations are the post-sanctions counterparts of the baseline labor-market-clearing and income conditions.

Counterfactual equilibrium. For given trade-cost changes, we solve the counterfactual equilibrium in relative changes using the exact-hat-algebra approach popularized by [Dekle et al. \(2007\)](#). Let x' denote the counterfactual value of variable x , and let $\hat{x} := x'/x$. The final trade-cost change entering the gravity equation is $\hat{\tau}_{ei}^k = \tau_{ei}^{k'}/\tau_{ei}^k$.

Given wages and final trade costs, the product-level price changes are

$$\hat{P}_i^k = \left[\sum_{e=1}^N \pi_{ei}^k (\hat{w}_e \hat{\tau}_{ei}^k)^{-\theta^k} \right]^{-1/\theta^k}. \quad (21)$$

The corresponding group-price changes are

$$\hat{P}_i^g = \left[\sum_{k \in \mathcal{K}_g} \frac{X_i^k}{X_i^g} (\hat{P}_i^k)^{1-\sigma_g} \right]^{1/(1-\sigma_g)}, \quad (22)$$

and the aggregate price-index change is

$$\hat{P}_i = \left[\sum_{g \in \mathcal{G}} \frac{X_i^g}{X_i} (\hat{P}_i^g)^{1-\nu} \right]^{1/(1-\nu)}. \quad (23)$$

Counterfactual product-level sourcing shares are

$$(\pi_{ei}^k)' = \frac{(\hat{w}_e \hat{\tau}_{ei}^k)^{-\theta^k} \pi_{ei}^k}{\sum_{\ell=1}^N (\hat{w}_\ell \hat{\tau}_{\ell i}^k)^{-\theta^k} \pi_{\ell i}^k}. \quad (24)$$

Our nested demand system determines counterfactual product expenditure:

$$X_i^{k'} = \hat{X}_i X_i^k \left(\frac{\hat{P}_i^{g(k)}}{\hat{P}_i} \right)^{1-\nu} \left(\frac{\hat{P}_i^k}{\hat{P}_i^{g(k)}} \right)^{1-\sigma_{g(k)}}. \quad (25)$$

Bilateral expenditure is then

$$X_{ei}^{k'} = (\pi_{ei}^k)' X_i^{k'}. \quad (26)$$

Using the baseline identity $X_e = w_e L_e$, equations (19) and (20) can be written in change form as

$$\hat{w}_e X_e = \sum_{i=1}^N \sum_{k=1}^K \frac{X_{ei}^{k'}}{\kappa_{ei}^k}, \quad (27)$$

$$\hat{X}_i = \hat{w}_i + \frac{R_i'}{X_i}. \quad (28)$$

The counterfactual equilibrium consists of wage changes $\{\hat{w}_i\}_{i=1}^N$ and expenditure changes $\{\hat{X}_i\}_{i=1}^N$ that jointly satisfy equations (18) and (21)–(28), together with a numéraire normalization. These changes determine product and group prices, trade shares, bilateral expenditure, and hub income. We solve for the equilibrium using a tâtonnement algorithm (see also [Alvarez and Lucas, 2007](#)).

Real income changes are given by the standard expression $\hat{W}_i = \frac{\hat{X}_i}{\hat{P}_i}$. Moreover, the nested demand structure also delivers group-specific real absorption. Let $X_i^{g'} = \sum_{k \in \mathcal{K}_g} X_i^{k'}$. Then

$$\hat{Q}_i^g = \frac{\hat{X}_i^g}{\hat{P}_i^g} = \hat{W}_i \left(\frac{\hat{P}_i}{\hat{P}_i^g} \right)^\nu. \quad (29)$$

In particular, $\hat{Q}_R^M$ and $\hat{Q}_R^C$ measure changes in Russia's real access to military-relevant and other critical goods.

Note that when setting $\sigma_g = \nu$ for all groups, $\rho_h = 1 \forall h$, and $\beta = 0$, the framework nests the standard multi-product EK model with one CES nest and no special rerouting technology (beyond the standard TI). We next explain how we bring the model to the data.

5.3 Bringing the Model to the Data

We bring the model to the data in two stages. First, using 2019 data, we construct the pre-sanction economy: bilateral trade flows, total, group-, and product-level expenditure, and a TI-consistent matrix of bilateral trade-cost levels. Second, we calibrate the sanctions-specific transshipment technology. The distribution of hubs observed in the 2024 Russian customs data disciplines their relative comparative advantages, while the triple-difference estimate in Section 4 disciplines the overall cost of rerouting. Thus, post-sanction outcomes enter the parameter calibration only through this final stage via ρ_h .²³

We calibrate the model at the 6-digit HS level, the most detailed product classification that is harmonized across countries. This disaggregation level comes with a data challenge, as explained below, but allows us to overcome several drawbacks of more aggregated quantitative exercises that are standard in the literature. First, export bans are defined at the product level and cannot be mapped cleanly into aggregate sectors. Second, the model predicts rerouting at the origin-product level, which can be compared directly with the customs data. Third, product-specific import shares and trade elasticities allow the ease of replacing a sanctioned origin to vary across narrowly defined products.

Trade and absorption data. We use data from 2019, the last full year before COVID and the war. Bilateral product-level trade flows from UN Comtrade measure X_{ei}^k for $e \neq i$, with products defined at the HS6 level in the HS2017 vintage. For each reporting side, we first aggregate records that are split by customs procedure, transport mode, and second

²³ The country-selection rule also uses the 2024 customs data to ensure that empirically important hubs are represented in the model's country sample. All baseline flows, expenditures, elasticities, and trade-cost levels are constructed from 2019 or other pre-sanction sources.

partner country. To limit the effect of missing or incomplete mirror reports, we then use the larger of the importer’s reported imports and the exporter’s reported exports.

To measure total expenditure, X_i , and the domestic expenditure underlying X_i^g , we use the OECD Inter-Country Input-Output tables ([regular ICIO, 2025 edition](#)). For each of the ICIO’s goods industries, expenditure on domestically produced goods equals gross output minus exports (set to zero where recorded exports exceed recorded output), and total expenditure adds imports to this amount. For countries missing from ICIO, we use the International Trade and Production Database for Estimation (ITPD-E, [Release 2](#); see [Borchert et al. \(2021\)](#)).²⁴ Moreover, to reduce the dimensionality of the quantification, given that we have over 5,000 products in the final sample, we keep a subset of countries from the UN Comtrade data. We restrict the sample to countries that (i) implement export bans against Russia,²⁵ (ii) are among Russia’s top 50 import origins or top 50 export destinations in 2019, (iii) are members of the Eurasian Economic Union, or (iv) act as transshipment hubs, defined as accounting for at least 0.5% of the 2024 customs value that reaches Russia through non-implementing countries for fully banned products from the EU (and mirroring origins), so that the sample contains the countries through which evasion runs. This leaves a total of 74 countries, which make up 96% (98%) of Russia’s total exports (imports) and 93% (92%) of world exports (imports).²⁶ Finally, we drop HS6 products for which Russia’s total 2019 trade is less than USD 10,000 on both the import and the export side. A further 11 products trade in too few markets for the regression below to deliver their product-level effects, leaving 5136 products in the final sample.²⁷

²⁴ The ITPD-E data sometimes imply very small (or even negative) expenditure on domestically produced goods. We therefore use these data only for countries where ITPD-E reports positive domestic expenditure in at least 50 of its 153 goods industries, which are more disaggregated than the ICIO industries, and these industries cover at least 40% of the country’s recorded imports. Otherwise, we use ICIO countries whose population lies on the same side of the sample median as a reference group, allowing home-expenditure shares to differ with country size. For each reference country, the home-expenditure share is expenditure on domestically produced goods divided by the sum of this expenditure and imports, and we take the unweighted average across reference countries. We choose the recipient country’s total domestic expenditure to reproduce that average share given its own imports, and distribute the total across industries according to domestic expenditure pooled across the reference group.

²⁵ Taiwan enters the sample as a sanctions implementer. Because partner countries record their trade with Taiwan only under the residual category “Other Asia, not elsewhere specified,” which may bundle other unassigned flows, partner-reported mirrors are not a reliable measure of Taiwanese trade. We therefore take Taiwan’s trade exclusively from its own customs reports to UN Comtrade (reporter code 490).

²⁶ We treat China and Hong Kong as a single country when calibrating the model (reassigning Hong Kong’s flows to China) due to large shares of re-exports between these countries, as is frequently done in the quantitative trade literature. We treat Liechtenstein and Switzerland in the same way, as Liechtenstein has no separate customs territory and no separate presence in UN Comtrade. In both cases, we combine the two territories’ flows on each reporting side before taking the maximum across sides. We do not aggregate the remaining countries into a rest-of-the-world region.

²⁷ Concretely, these are the products for which the regression returns no estimated importer-product effect or no estimated exporter-product effect for any country, after singleton observations are removed.

Elasticities. As is typically done in the literature, the elasticity values are taken off the shelf. We take θ^k from Fontangé et al. (2022), who estimate product-level trade elasticities using gravity regressions consistent with our setup above. For 6-digit products that cannot be matched to their data, we use the average elasticity within the product’s 4-digit heading. Moreover, due to some outliers in the elasticity data, we winsorize by bounding values below the first and above the 99th percentiles. The 85 products without a match at either level are assigned the median of the matched elasticities. The resulting minimum, median, and maximum of θ^k are, respectively, 1.5, 8.2, and 44.1. Every θ^k exceeds $\sigma_g - 1$ (explained next), which ensures that the exponent used in the expenditure recovery below is positive and that one can choose an η^k satisfying both $\eta^k > \sigma_g$ and $\theta^k > \eta^k - 1$ (see Section 5.1). Because η^k otherwise affects only the product-specific price-index constant that we absorb into T_e^k , it need not be calibrated separately, a standard result in EK-type models.

Within groups, we set $\sigma_g = 2.2$ for all three groups, in line with French (2016). Across groups, we set $\nu = 1.1$, close to the Cobb–Douglas case and half the within-group elasticity, so that expenditure shifts across the three broad groups are relatively limited. The robustness checks below consider values of ν below one as well as the flat-CES case $\nu = \sigma_g$. Our main results barely move with these elasticities.

Product groups. The assignment of the 5136 products to M , C , and O follows published lists. Group M starts from the 44 battlefield-relevant products listed in Table A1 of Scheckenhofer et al. (2025) and, after two adjustments, contains 42 products.²⁸ Table A1 predates the February 2024 update of the Common High Priority Items list (version of February 2024), which added five machine-tool products (Tier 4.B). These are covered by the two lists introduced next and therefore fall into C . Beyond battlefield goods, sanctions target products with high strategic value. The EU’s export-control framework identifies this set through two lists: the dual-use list, covering goods with both civilian and military uses, and the advanced-technology list (Annex VII), covering goods and technology that might contribute to Russia’s military and technological enhancement. Group C contains the products that appear in either list, spanning machine tools, electronics, precision instruments, chemicals, and metals. Both lists are defined below the HS6 level, and we

²⁸ First, we move the three products of the HS 8517 phone family (851712, 851762, and 851769) to C , because they entered the battlefield list as sources of components rather than as military equipment. Second, we add HS 854150 (other semiconductor devices): the battlefield list is published in a later HS vintage, in which 854150 is split into two successor codes, and both are on the list.

assign an HS6 product to C if any of its lines appears on either list.²⁹ Net of the products already in M , this yields 967 products.³⁰ The remaining 4127 products form group O .

The partition also enters the expenditure construction, because the group tier of demand requires the expenditure levels X_i^g . For imports this is immediate, as every product belongs to exactly one group, so import expenditure sums directly to group totals. However, domestic absorption is recorded by ICIO industry rather than by product, and we allocate it in two steps. First, we map each HS6 product to one of the ICIO’s 26 goods industries (agriculture, mining, and manufacturing) using all rows of the official [UN correspondence tables](#) (HS2017 to CPC 2.1 to ISIC Rev. 4), including those flagged as partial. For 95.6% of import value in our sample, these rows single out one industry.³¹ Second, we split each industry’s domestic absorption across M , C , and O in the same proportions as the country’s imports within that industry, since imports are the only expenditure we observe at the product level. Exactly one country-industry cell has no imports. For that cell we use the industry’s import composition summed over all sample countries. Group imports plus allocated domestic absorption then give X_i^g .³² This allocation assumes that, within an ICIO industry, the $M/C/O$ composition of expenditure on domestically produced goods is the same as the composition of imports. For Russia, the three groups account for 1.6% (M), 30.7% (C), and 67.8% (O) of total expenditure.

²⁹ One listed line suffices here, because the groups describe what a product is, while the ban classification in Section 2.2 requires all of a product’s lines to be banned (“full” bans) so as not to overstate the shock.

³⁰ The product lines come from the European Commission’s correlation tables. For the dual-use list (Regulation 2021/821), we use the table valid through 31 December 2024 (available [here](#); retrieved 4 August 2026), matching our end-2024 sanctions data, while for Annex VII (Regulation 833/2014), the 2026 vintage (available [here](#); retrieved 28 July 2026) is the earliest available to us. The dual-use table lists ten-digit TARIC codes, which we truncate to the six-digit level, expanding the few entries given only at the four-digit heading level to all subheadings of that heading. From the Annex VII table we take every complete eight-digit CN code, whereas rows without one (software and technology items, free text) identify no product line and are dropped. Codes published in later HS vintages are converted to HS2017 with the one-to-one conversion sheet of the official [UN conversion table](#).

³¹ For the remaining 4.4% of import value, the correspondence tables give no industry or several. We then assign the ICIO industry that produces the goods of the product’s HS2 chapter under the industry definitions (chapters 28 and 29 to chemicals, for example). Where the tables offer several candidate industries and the chapter industry is not among them, we take the candidate to which the largest number of unambiguously mapped products in the same HS4 heading belong. For the four products where neither rule decides (HS 310100, 440210, 440290, and 940690), we take the candidate that comes first in the ICIO industry code order.

³² Writing D_{ir} for domestic absorption and I_{ir}^g for imports of group g in country i and ICIO industry r , the allocation is

$$D_{ir}^g = \begin{cases} D_{ir} \frac{I_{ir}^g}{\sum_s I_{ir}^s}, & \text{if } \sum_s I_{ir}^s > 0, \\ D_{ir} \frac{\sum_j I_{jr}^g}{\sum_j \sum_s I_{jr}^s}, & \text{if } \sum_s I_{ir}^s = 0, \end{cases}$$

where s runs over the three groups and j over the sample countries, with $X_i^g = \sum_r (D_{ir}^g + I_{ir}^g)$.

Product-level expenditure. We cannot read X_i^k (i.e., product-level expenditure) directly from the data, because expenditure is not observable at this level of disaggregation globally. We therefore recover k -level expenditure (and, accordingly, domestic product-level trade) using the model's structure.³³

The preceding step delivers group expenditure, X_i^g , so what remains is to allocate it across the products of each group. By equation (9), this allocation is pinned down by the product-level price indices, P_i^k , which we need to estimate. We cannot simply use equation (11) to estimate P_i^k , because the dependent variable itself is scaled by X_i^k . However, combining equations (9), (11), and (12), we obtain

$$\frac{X_{ei}^k}{X_{ei}} = \frac{T_e^k w_e^{-\theta^k} \Lambda_i^{g(k)} (P_i^k)^{\theta^k - (\sigma_{g(k)} - 1)} (\tau_{ei}^k)^{-\theta^k}}{\sum_{s=1}^K T_e^s w_e^{-\theta^s} \Lambda_i^{g(s)} (P_i^s)^{\theta^s - (\sigma_{g(s)} - 1)} (\tau_{ei}^s)^{-\theta^s}},$$

where $X_{ei} := \sum_{s=1}^K X_{ei}^s$ denotes total bilateral trade and $\Lambda_i^g := X_i^g (P_i^g)^{\sigma_g - 1}$ collects the group-level terms, which are common to all products of group g in country i . Taking logs of this equation yields

$$\log \left(\frac{X_{ei}^k}{X_{ei}} \right) = \log \left(T_e^k w_e^{-\theta^k} \right) + \log \left(\Lambda_i^{g(k)} (P_i^k)^{\theta^k - (\sigma_{g(k)} - 1)} \right) - \log (\Psi_{ei}) - \theta^k \log (\tau_{ei}^k),$$

where $\Psi_{ei} := \sum_{s=1}^K T_e^s w_e^{-\theta^s} \Lambda_i^{g(s)} (P_i^s)^{\theta^s - (\sigma_{g(s)} - 1)} (\tau_{ei}^s)^{-\theta^s}$. We run an OLS regression with $\log (X_{ei}^k / X_{ei})$ as the dependent variable on all positive bilateral flows. The first three elements on the right-hand side can be captured by an exporter-product fixed effect (FE), an importer-product FE, and an exporter-importer FE, respectively. To parametrize the impact of τ_{ei}^k , we interact a set of product dummies with several standard bilateral variables: the log of population-weighted distance, as well as dummy variables indicating whether the country pair has a free trade agreement or is part of the same customs union, and whether it shares a common border, language, or colonial history. These control variables are obtained from the standard [CEPII Gravity database](#) (Conte et al., 2022).

Therefore, through the lens of the model, the exponentiated importer-product FE yields the composite $B_i^k := \Lambda_i^{g(k)} (P_i^k)^{\theta^k - (\sigma_{g(k)} - 1)}$ rather than the price index alone. Because every θ^k exceeds $\sigma_g - 1$, this exponent is positive, so a higher B_i^k corresponds to a higher P_i^k at a given group scale. The regression pins down the exporter-product and importer-product FEs only jointly, so a product-specific constant can be shifted from one set of effects to the other without changing any fitted share. We normalize the estimated B_i^k to a geometric mean of one across importers within each product, absorbing the offsetting constant into the exporter-product effects, so that all fitted values are unchanged. For importer-product

³³ French (2016) recovers product-level expenditure while assuming that trade costs and trade elasticities are common across products, whereas Torun (2025) uses a non-homothetic final-goods aggregator to do so.

cells where the regression delivers no estimate, we set B_i^k equal to the maximum estimate within k (recall that a higher P_i^k implies more difficult access to cheap sources). Moreover, due to outliers in these estimates, we winsorize B_i^k by bounding values below the 10th and above the 90th percentiles within each importer i and group g . Recovering P_i^k from B_i^k then requires the group scales Λ_i^g , which the regression does not deliver. We normalize $\Lambda_i^O = 1$ and solve, country by country, for the two remaining scales such that the implied group expenditures match their measured ratios (with X_i^g measured as described above),

$$\frac{\Lambda_i^g (P_i^g)^{1-\sigma_g}}{(P_i^O)^{1-\sigma_O}} = \frac{X_i^g}{X_i^O}, \quad g \in \{M, C\},$$

where the group price indices follow from equation (6) and $\Lambda_i^g (P_i^g)^{1-\sigma_g}$ is the group expenditure for a given scale. The left-hand side is strictly increasing in Λ_i^g , so each equation has a unique solution. Equation (9) then yields X_i^k , which adds up to X_i^g within each group, and hence to X_i , by construction. Domestic product-level trade follows as the residual $X_i^k - \sum_{e \neq i} X_{ei}^k$. The elasticity ν plays no role in this step, because group expenditure is already measured at this point, but it governs how expenditure shifts across the three groups in the counterfactuals.

Because product-level expenditure is recovered rather than directly observed, some country-product cells imply very small or even negative domestic expenditure relative to observed imports. As a final step, we therefore impose a 10% lower bound on each product's domestic expenditure share. We impose the bound with two rescalings within each group. First, wherever the home share falls below the bound, we raise X_i^k until domestic expenditure is exactly 10% of total product expenditure. The bound initially binds for 15% of all country-product cells. Second, we scale down the expenditure of the group's remaining products proportionally, so that group expenditure (and hence total expenditure) is unchanged, and we repeat both steps until no product violates the bound. Observed imports are never altered. For 19 of the 74 countries, the bound requires more expenditure in at least one group than that group's measured total. There we raise the group's expenditure share to the smallest value consistent with the bound and scale the other groups' shares down proportionally, which changes a group's share by at most 1.8 pp and again leaves X_i unchanged. Importantly, Russia is not among these countries. The resulting domestic and bilateral flows give the baseline trade shares, π_{ei}^k . The robustness analysis reports results under a lower domestic-share bound.

Baseline trade costs. Once a counterfactual trade-cost change ($\hat{\tau}$) is specified, the exact-hat system can be solved using baseline trade shares, as described in Section 5.2.3. The transshipment mechanism additionally requires bilateral trade-cost *levels*, because identifying the cheapest intermediary requires comparing alternative routes to Russia.

We recover these levels in two steps. We first invert the gravity equation to obtain the bilateral trade barriers that rationalize the observed baseline trade shares and then use the triangle inequalities to recover the TI-consistent baseline trade-cost matrix used to compare routes after the export bans.

To recover the initial trade barriers, we use the trade shares, π_{ei}^k , and the exporter-product FEs of the regression above, whose exponential yields $T_e^k w_e^{-\theta^k}$. Similar to the above, for exporter-product cells where the regression delivers no estimate, we set $T_e^k w_e^{-\theta^k}$ equal to the minimum estimate within k (recall that a lower $T_e^k w_e^{-\theta^k}$ implies less export competitiveness), and we winsorize by bounding values below the 10th and above the 90th percentiles within each exporter e across all products. The winsorization of B_i^k is done within group instead, because B_i^k contains the group scale Λ_i^g , whose level differs across groups. According to the gravity equation, trade barriers can be backed out using bilateral relative to domestic flows, that is,

$$\tau_{ei}^k = \left(\frac{\pi_{ei}^k / (T_e^k w_e^{-\theta^k})}{\pi_{ii}^k / (T_i^k w_i^{-\theta^k})} \right)^{-1/\theta^k}, \quad (30)$$

which will match trade flows exactly, in line with the hat-algebra logic.³⁴

We then need to ensure that the initial trade costs satisfy the TIs to have a theory-consistent starting point. The gravity-implied barriers from equation (30) need not satisfy the triangle inequalities, which can arise because the bilateral term recovered from gravity combines route-related trade costs with other pair-specific forces, such as bilateral preference shifters (Foellmi et al., 2024). To recover trade costs, we apply the algorithm from Foellmi et al. (2024, Section 4) and update the τ_{ei}^k values from equation (30) iteratively until we have no more TI violations. This yields a set of TI-consistent trade costs, which we then use to measure transshipment opportunities following a shock.³⁵

Post-sanction hub wedges. The preceding steps complete the calibration of the 2019 baseline economy, including the TI-consistent bilateral trade-cost matrix. We now turn to the hub wedges, ρ_h , which describe the sanctions-specific transshipment technology

³⁴ For zero bilateral trade shares, we replace the relative share term in parentheses by 10^{-12} , so the gravity-implied barrier passed to the TI algorithm is $\tau_{ei}^k = 10^{12/\theta^k}$.

³⁵ The algorithm starts by setting the trade barriers that we obtain from equation (30) to at least one. Then, it verifies the TIs, and if a TI is violated, it sets the updated trade cost equal to the minimum indirect cost. This procedure is repeated until there are no more TI violations. The wedge between the initial value of trade barriers and the final value of trade costs can be interpreted as a bilateral preference shifter through the lens of standard gravity theory (cf. Foellmi et al., 2024). Because exact hat algebra conditions on the observed baseline trade shares, this residual need not be recovered separately. The levels τ_{ei}^k are needed only to determine rerouting opportunities and the corresponding trade-cost changes.

introduced in Section 5.2.2. These are the only model parameters calibrated directly to post-sanction outcomes.

It is useful to separate relative hub attractiveness, which determines hub choice for given trade costs, from the overall cost of evasion. Write

$$\rho_h = a\tilde{\rho}_h, \quad \frac{1}{H} \sum_{h=1}^H \log \tilde{\rho}_h = 0,$$

so that the relative wedges $\tilde{\rho}_h$ have a geometric mean of one. Their relative values determine which intermediaries are selected, while the common scale a governs the overall cost of rerouting. We discipline the former using the distribution of hubs observed in the 2024 Russian customs data and the latter using the triple-difference-implied trade response.

Relative wedges. We use the rerouting via non-coalition dispatch countries observed in the 2024 Russian customs data for fully banned products from EU origins and countries that mirror the EU sanctions. We restrict the calibration to these origins because our sanctions data identify their full product-level bans precisely. For each potential intermediary, the empirical target is its share of positive origin-product-hub relations, renormalized over the countries represented in the model sample. Figure 9 first shows the model’s routing when all relative wedges are equal. Each banned origin-product cell is then assigned to the eligible intermediary with the lowest TI-based two-leg cost. The pre-war trade network already identifies the main hubs. In both the data and the model, transshipment is highly concentrated among a small number of countries. The five largest hubs in the data are Belarus, China, Serbia, Türkiye, and the United Arab Emirates, while India replaces the United Arab Emirates among the model’s five largest hubs. Their combined shares are 82% in the data and 64% in the model. The correlation across all hubs is 0.76.

Note that this is a demanding untargeted pre-calibration comparison. The route costs use 2019 trade data and do not use the observed hub shares. At the individual-hub level, the TI benchmark understates the roles of Türkiye and the United Arab Emirates and overstates those of Belarus, India, and China. We then choose the relative wedges, $\{\tilde{\rho}_h\}_{h=1}^H$, to minimize the distance between the model’s and the observed hub-share distributions. With the calibrated wedges, the model closely reproduces the targeted aggregate shares.³⁶

³⁶ The calibration universe contains 25 411 fully banned origin-product cells from EU origins and countries that mirror the EU sanctions, each with positive 2019 imports to Russia. Within this universe, the empirical hub shares are constructed from positive 2024 origin-product-hub relations and renormalized over the intermediaries represented in the model. Denoting the model’s and the data’s hub-use shares by s_h and s_h^{data} , respectively, the relative wedges solve

$$\min_{\{\tilde{\rho}_h\}} \left\{ \sum_h (s_h - s_h^{\text{data}})^2 + \lambda \sum_h (\log \tilde{\rho}_h)^2 \right\}$$

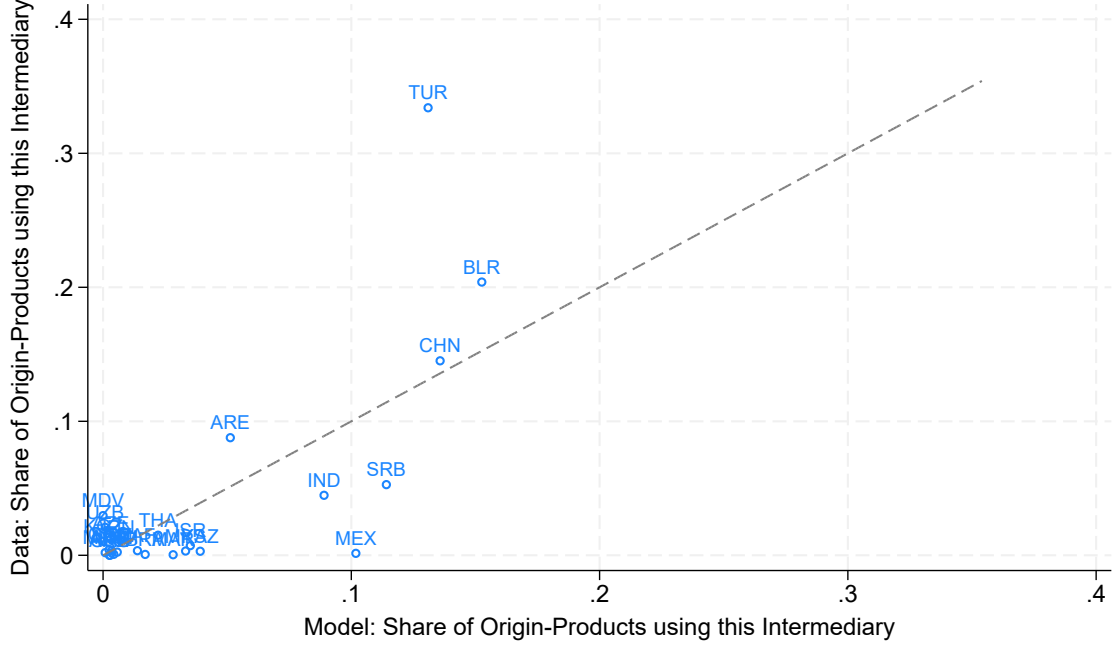


FIGURE 9. TRANSSHIPMENT HUBS: BENCHMARK PREDICTION VS. DATA

Note: This figure compares the hub shares in the 2024 Russian customs data with the hubs selected by the model after the direct export bans when $\rho_h = 1$ for all h . The calibration universe consists of fully banned origin-product cells from EU origins and countries that mirror the EU sanctions, with positive 2019 exports to Russia. Empirical shares are constructed from positive routes through non-coalition dispatch countries and renormalized over dispatch countries represented in the model. A hub's empirical share is the number of positive origin-product-hub relations involving that hub divided by the total number of such relations. In the model, each origin-product cell contributes one relation because it selects a single minimum-cost hub. The dashed line is a 45°-line.

Common scale. The hub-share distribution identifies relative hub comparative advantage but not its common scale. If the lower bound in equation (15) were never binding, multiplying all relative route costs by the same factor would leave the ranking of hubs and the route-competition component of the markups unchanged, while changing only the overall cost of rerouting. Because the lower bound can bind, this invariance is only approximate, as the common scale can also affect route choices and markups.

The static triple-difference estimate in Section 4, , implies that a fully banned flow falls to approximately 15% of its counterfactual level. Through the gravity equation, the corresponding partial-equilibrium response for origin-product cell (e, k) is $(\tau_{eR}^{k'}/\tau_{eR}^k)^{-\theta^k}$, where $\tau_{eR}^{k'}$ is the final importer-facing cost of the selected route, including routing costs and its markup. We choose the common scale a so that the 2019-trade-weighted mean of this response across the calibration cells equals $\exp()$. At each candidate value of a , we recompute the selected routes and their markups. Because the lower bound can also affect the relative route costs, we iterate between the relative wedges and the common scale

subject to $\sum_h \log \tilde{\rho}_h = 0$, with $\lambda = 10^{-4}$. The penalty shrinks weakly identified relative wedges toward the common value and matters primarily for countries that intermediate little in both the model and the data.

until both are stable. Under the geometric-mean normalization of $\tilde{\rho}_h$, the calibrated scale is 1.34. The economically relevant objects, however, are the final hub wedges, $\rho_h = a\tilde{\rho}_h$, and the resulting route-level trade costs. The implied increases in delivered costs are highly heterogeneous across the calibration cells. For the median banned flow, the effective trade cost rises by 69%. In the counterfactuals, the calibrated wedges apply to all banned coalition-origin flows.

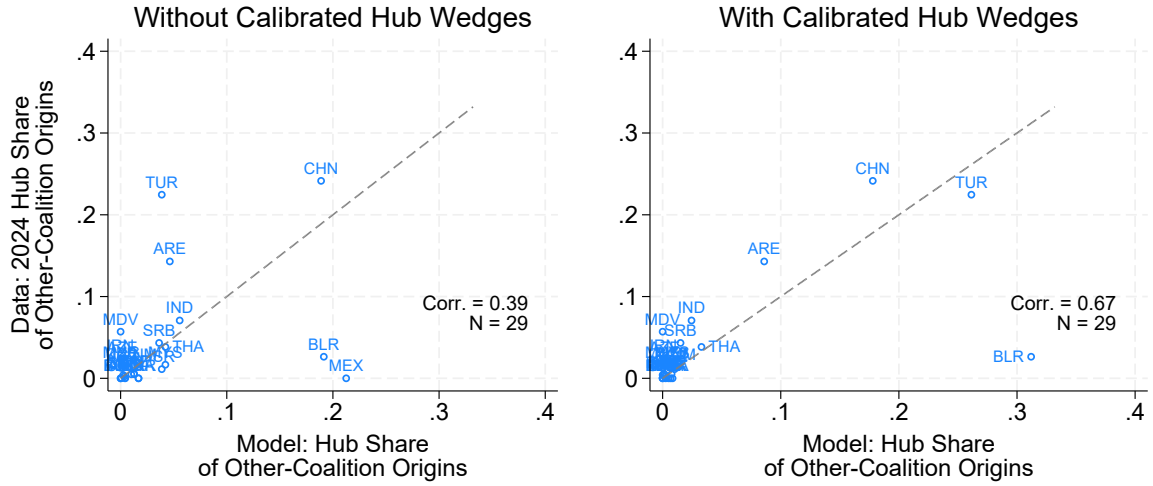
Untargeted fit. Only the aggregate hub-share vector for EU origins and countries that mirror the EU sanctions enters the calibration of the relative wedges. The routing of the remaining coalition origins, the distribution of hubs within individual origins, the value-weighted hub shares, and the variation across product sectors are therefore untargeted. Figure 10 reports the first two comparisons. For coalition origins outside the calibration sample, the correlation between the model and the data across hubs rises substantially from 0.39 without the calibrated wedges to 0.67 with them. Across origin-hub pairs (considering the entire coalition), the corresponding correlation rises from 0.55 to 0.90. Appendix Figure B.3 shows analogous improvements for value-weighted hub shares and for shares within HS2 chapters. These comparisons use the same 2024 customs source as the calibration target and are therefore untargeted rather than out-of-sample. They show that a single set of hub-specific wedges fitted to the aggregate extensive-margin shares also improves the fit of substantially more disaggregated routing patterns.

6 The Aggregate Effects of Export Bans and Transshipment

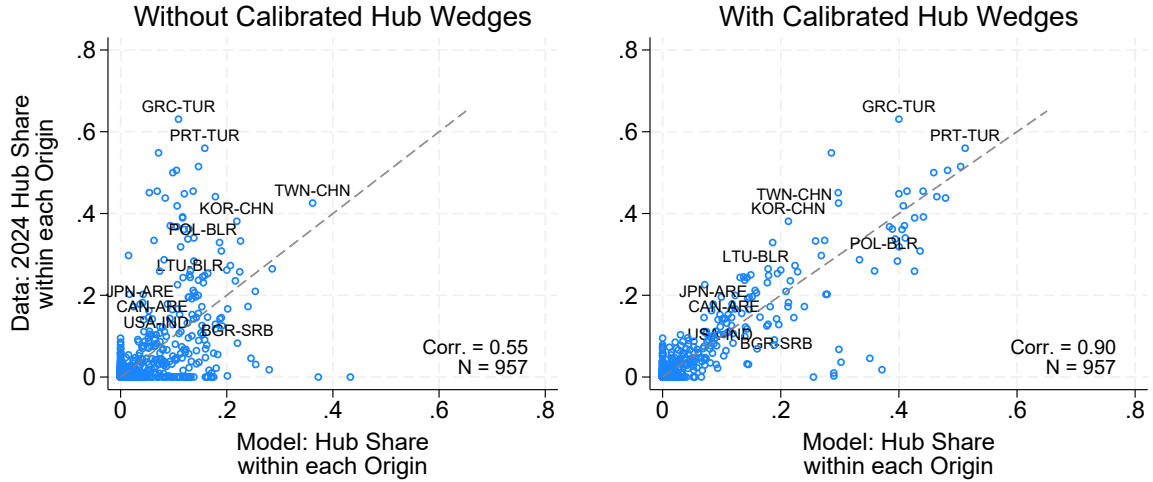
This section asks how much the current export bans reduce Russia’s imports and real income once transshipment is taken into account, and how much further full enforcement would go. We compare two post-ban equilibria. Under observed evasion, banned coalition-origin goods can be rerouted through the calibrated transshipment technology. Under full enforcement, all such indirect routes are blocked. The comparison isolates the *enforcement gap*, i.e., the part of the full-enforcement real-income loss that does not materialize while transshipment remains possible.

We apply the origin-product export bans in force by the end of 2024, accounting for differences in product coverage across coalition members. Coalition countries cannot serve as intermediaries, including for products they do not themselves ban.³⁷ The export bans that the EU imposed on Belarus are fully enforced throughout, such that it can never be used as a hub for these products (see Section 5.3 for details).

³⁷ This treats coalition membership as a commitment not to facilitate the circumvention of another member’s export bans. We treat Ukraine as part of the coalition but assume that its export restrictions are fully enforced and that it never acts as a transshipment hub.



(A) HUB SHARES OF THE OTHER-COALITION ORIGINS



(B) HUB SHARES WITHIN EACH ORIGIN, BY ORIGIN-HUB PAIR

FIGURE 10. UNTARGETED FIT OF THE CALIBRATED HUB WEDGES

Note: This figure compares the model's routing of banned flows with the routing observed in the 2024 Russian customs data, without the calibrated hub wedges in the left panels and with them in the right panels. The matched sample contains origin-product cells for which the model reroutes a banned flow and the customs data record at least one positive route, restricted to the model's 74 sample countries; routes dispatched through countries outside the model sample are excluded. The sample contains 6961 matched origin-product cells across all coalition origins. In the data, a cell may contribute positive relations under several hubs, whereas in the model, it contributes one relation under the selected hub. Panel 10a aggregates these relations by hub for coalition origins outside the EU-plus-mirroring calibration group. Panel 10b computes hub shares separately within each coalition origin and shows the 33 origins with at least 25 matched cells. Shares sum to one within each origin. For legibility, labels identify ten high-share origin-hub pairs, with at most two labels per hub. The dashed line is a 45°-line. "Corr." is the unweighted Pearson correlation across the plotted points, and "N" is the number of plotted points.

All scenarios start from the same theory-consistent 2019 baseline, where we first solve the equilibrium to have balanced trade in the baseline levels. Because nominal flows are

identified only relative to the model’s numéraire, we report import effects as percentage changes in the corresponding flows’ shares of world expenditure. This measure is invariant to the normalization of nominal values. Table 3 reports the main results for Russia. We turn to the effects on EU exporters below.

The enforcement gap. Table 3 gives the main quantitative result. Under observed evasion, the export bans reduce Russia’s real income by 1.26% relative to the no-sanctions baseline. A loss of this size is substantial by the standards of quantitative trade models.³⁸ Under full enforcement, the loss rises to 1.66%. We call the absolute difference between these two losses the *enforcement gap*. Relative to the observed-evasion loss, full enforcement raises Russia’s loss by 32%. Relative to the full-enforcement loss, observed evasion offsets 24% of the loss under full enforcement. Although real income changes correspond to welfare changes in our model, sanctions are typically motivated by reasons beyond income effects, which is why we will explicitly call them real income losses throughout.

The enforcement gap also has a money-metric interpretation. With homothetic preferences, the difference in equivalent variation between observed transshipment (OT) and no transshipment (NT), evaluated at baseline prices, is $X_R (\hat{W}_R^{\text{OT}} - \hat{W}_R^{\text{NT}})$, where X_R is Russia’s annual baseline expenditure. This amounts to approximately USD 4.5 billion per year in 2019-base dollars. It is a welfare-equivalent amount rather than a fiscal transfer or a measure of resources directly available to the Russian state. However, to put the scale into perspective, this amount is numerically equivalent to the reported cost of about 130,000 Russian-produced strike drones, or to the first-year compensation packages of about 70,000 contract soldiers recruited in Moscow.³⁹

Trade flows and strategic access. The trade responses explain why the sanctions can eliminate most targeted flows without producing a comparable fall in Russia’s overall access to goods. Under observed evasion, imports of banned coalition-origin products fall by 76%. Rerouted products retain their coalition origin in the model, so this decline is already net of evasion. It is also close to the decline observed in the customs data in 2024

³⁸ For comparison, using a related framework, Caliendo and Parro (2015) estimate that NAFTA’s tariff reductions raised U.S. welfare by 0.08%. See Costinot and Rodríguez-Clare (2014) for an overview of welfare magnitudes in quantitative trade models.

³⁹ We value drones at USD 35,000, the midpoint of reported unit costs of USD 20,000 to 50,000 for Russian-produced Shahed-type strike drones (CSIS, May 2025; Kyiv Independent, May 2025), and soldiers at the USD 60,000 of first-year pay that Reuters (July 2024) reports for contract soldiers recruited in Moscow. These benchmark prices are quoted in 2024–2025 dollars, whereas the welfare equivalent is expressed in 2019-base dollars. We leave them unadjusted, but a common-dollar inflation adjustment would raise the implied counts, so the reported equivalents are conservative in this sense.

TABLE 3. THE AGGREGATE EFFECTS OF THE SANCTIONS ON RUSSIA

Scenario	Change in Imports			Change in Real Income, $\hat{W}$	
	Full	Partial	Total	Change	vs. Observed Transshipment
Observed Transshipment	-76%	-27%	-9.1%	-1.26%	
No Transshipment	-100%	-24%	-11.1%	-1.66%	+32%

Notes. This table reports how Russia’s imports and real income change relative to the baseline without the sanctions. The first import column refers to Russia’s imports of banned products from coalition countries, the second to Russia’s total imports. Import entries are percentage changes in the corresponding flows’ shares of world expenditure. Real income changes are reported in percent, that is, $(\hat{W} - 1) \times 100$, and the last column reports the percentage increase in Russia’s real-income loss under full enforcement, relative to the observed-evasion loss. Under “Observed Evasion,” banned coalition exports can be rerouted through third-country hubs at the calibrated route costs and markups. Under “Full Enforcement,” rerouting is blocked, so there are no coalition exports of banned products and the -100% entries are mechanical. Ratios reported in the text are computed from unrounded model output. The results are discussed in Section 6.

(Figure 4), even though the calibration targets the average partial-equilibrium response of the banned flows rather than their general-equilibrium decline.⁴⁰

The banned flows account for 27% of Russia’s baseline imports. If no other import flow changed, their decline would reduce total imports by approximately 21%. Instead, total imports fall by only 9.1%, because Russia substitutes toward non-sanctioning origins and products that are not banned. Under full enforcement, the decline in the targeted coalition-origin flows rises (mechanically) from 76% to 100%, whereas the decline in total imports rises only from 9.1% to 11.1%.

The distinction is particularly important for military-relevant products. Banned coalition-origin flows account for 51% of Russia’s baseline imports in group M and fall by 91% under observed evasion. Total imports of military-relevant products fall by 12%, while real absorption of the broader military-relevant composite falls by 3.6%. The difference reflects several margins. The group composite includes domestic supply as well as imports from all origins. Russia can shift toward non-sanctioning or domestic suppliers, substitute across products within group M , and limit the reallocation of expenditure away from the military-relevant basket. Under full enforcement, the decline in military-relevant real absorption rises to 4.0%. The nested demand structure therefore matters for how the targeted-flow decline translates into strategic access (Appendix Table C.3), even though it has little effect on the aggregate enforcement gap (Appendix Table C.2).

Gains to third countries. Third countries can benefit from the sanctions through two distinct channels. They may sell more of their own products to Russia as trade is diverted

⁴⁰ The common scale of the hub wedges is chosen so that the 2019-trade-weighted average of the direct responses, $(\tau_{eR}^{k'}/\tau_{eR}^k)^{-\theta^k}$, matches the triple-difference moment. Before GE adjustments, this implies that the average banned flow retains 15.2% of its no-ban level (see Section 5.3).

away from coalition suppliers, and they may earn rents by transshipping coalition-origin goods. Under observed evasion, hubs receive USD billion in 2019-base rents, equal to of Russia’s baseline spending on the affected coalition flows.

These gains are concentrated. Relative to the no-sanctions baseline, real income rises by 0.01% in Türkiye, 0.04% in the United Arab Emirates, and 0.08% in Serbia. Appendix Table C.1 separates the direct contribution of intermediation rents from the real-wage component, which captures trade diversion and the remaining GE effects. Türkiye and the United Arab Emirates gain mainly through rents, Serbia through both margins, while China’s gain arises primarily through the expansion of its own exports. This distinction will matter in Section 7, where shutting down a country’s hub activity removes its intermediation role, whereas coalition accession also restricts its own exports. Recall that the rent estimates use our benchmark value $\beta = 1$, under which the selected hub can extract its full route advantage over the second-best hub (up to the demand-based cap). They should therefore be read as an upper-bound benchmark within our specification, rather than as point estimates of intermediary profits.

Impact on the EU. Table 4 reports the corresponding effects on EU-origin trade and EU real income. Under observed evasion, Russian expenditure on the affected EU-origin goods falls by 76%. Because the observed-evasion measure contains the markups retained by intermediary hubs, it does not correspond to the original producers’ net revenue.

The real-income effects are modest relative to Russia’s losses. Under observed evasion, the median member state loses 0.036% of real income, and the losses are concentrated among members with high pre-war exposure to Russia. Lithuania loses 0.46% and Estonia 0.33%, while the effects in Portugal and Spain are close to zero. A few members gain slightly through trade diversion (Ireland’s real income, for example, rises by -0.003%).

Full enforcement raises the median member state’s loss by 35%, somewhat larger than the corresponding 32% increase for Russia. The proportional effect of stronger enforcement is therefore of comparable size on the two sides, while the median member-state loss remains more than an order of magnitude smaller than Russia’s in levels.

Robustness. Appendix Table C.2 recomputes the main comparison under alternative parameter choices, recalibrating the relative hub wedges and their common scale to the same empirical moments in every specification. The main object of interest is the increase in the full-enforcement loss relative to the observed-evasion loss, which equals 32% in the benchmark. This is the enforcement gap expressed relative to the observed-evasion loss.

The demand-side elasticities, which are set rather than estimated, have little effect on this aggregate comparison. Varying the cross-group elasticity ν between 0.5 and

TABLE 4. THE EFFECTS OF THE SANCTIONS ON THE EU

Country	Full baseline USD M	Exports to Russia (%)						Real Income (%)		
		Full		Partial		Total				
		OT	NT	OT	NT	OT	NT	OT	NT	NT-OT (pp)
EU27	58,838	-76%	-100%	-27%	-23%	-39%	-48%	-0.036%	-0.048%	-0.009

Notes. This table reports changes in banned EU-origin flows to Russia and in EU real income under the observed-evasion and full-enforcement scenarios of Table 3. The first column reports the baseline value of the affected EU-origin flows in millions of 2019 USD. The flow changes are measured gross of the markups retained by intermediary hubs and therefore correspond to Russian expenditure on EU-origin goods rather than to the original exporters' net receipts. Under "Full Enf.," no affected EU-origin goods reach Russia, so the -100% entry is mechanical. Real-income changes refer to the median member-state change, so that the entry reflects a typical member state rather than the most exposed members. Appendix Table C.4 reports the member-state results.

2.2, where $\nu = 2.2$ imposes equally easy substitution across and within groups, and lowering the within-group elasticity for military-relevant products to $\sigma_M = 1.2$ leaves the enforcement gap between 32.3% and 32.4%. The nested structure matters more for how the sanctions affect the composition of Russian absorption than for the aggregate difference between observed and full enforcement. Appendix Table C.3 reports the group-level responses across the specifications. With lower cross-group substitutability, military-relevant absorption shrinks less and the burden shifts toward other goods. However, note that full enforcement deepens the military-relevant decline by a roughly constant proportion across the demand-side variants.

The product-level domestic-share floor matters more. Because domestic product expenditure is recovered rather than directly observed, the benchmark imposes a 10% lower bound on home shares. Lowering the bound to 5% widens the relative enforcement gap to 37%. Importantly, every specification is recalibrated, so the lower floor changes both the baseline trade shares and the fitted transshipment technology. In this case, the observed-evasion loss falls slightly while the full-enforcement loss rises. The 10% benchmark is therefore conservative with respect to the size of the enforcement gap.

The decomposition of the observed route wedge into rent and non-rent costs also has little effect on Russia's aggregate loss. Setting $\beta = 0$ eliminates hub rents and reallocates the fitted wedge toward the non-rent route-cost component when the model is recalibrated to the same empirical moments. The resulting relative enforcement gap is . Thus, Russia's aggregate response is governed mainly by the gross importer-facing route costs, while β determines where the associated income accrues.⁴¹

⁴¹ The real-wage component under the $\beta = 1$ benchmark is almost perfectly correlated with the total real-income change under the recalibrated $\beta = 0$ model, with a correlation of .

Wartime conditions may shift Russian expenditure toward military-relevant goods relative to the 2019 benchmark. Therefore, as a robustness check, we triple Russia’s demand weight for group M (i.e., $\hat{\alpha}_R^M = 3$) and first solve for the corresponding no-sanctions equilibrium. This raises the military-relevant group’s baseline expenditure share from 1.6% to 4.5%. We then impose the export bans relative to this demand-shocked reference economy, holding the calibrated transshipment technology fixed. The greater weight on military-relevant goods raises the real-income losses modestly under both observed evasion and full enforcement, to 1.32% and 1.72%, respectively. The relative enforcement gap remains 31%, close to the benchmark. Thus, a wartime shift toward military-relevant demand raises the level of the sanctions’ aggregate effect but does not materially change the relative transshipment gains.

Finally, the product-level trade elasticities are based on long-run gravity estimates, while the relevant response elasticity over the first years of the sanctions may be smaller. [Boehm et al. \(2023\)](#), for example, estimate a short-run elasticity that is less than half their long-run value. Halving our trade elasticities raises the loss under observed evasion sharply, to 2.29%, and widens the relative enforcement gap to 28%. This is expected, because when it becomes harder to substitute across source countries, a given increase in delivered costs generates a larger product-price response. The levels of the losses therefore rise substantially, while the relative importance of evasion remains of a similar magnitude.

Across the reported specifications, Russia’s imports of banned coalition-origin products fall by between 75% and 78%, while full enforcement raises the observed-evasion loss by between 32% and 37%. Transshipment therefore remains quantitatively important throughout the calibration variants above.

The scope of the current bans. Full enforcement is the most damaging implementation of the coalition’s *current* product restrictions, but it is not an upper bound on export sanctions more generally. If coalition countries instead stopped exporting every product to Russia and all rerouting were blocked, Russia’s real-income loss would reach 3.47% and its total imports would fall by 25.8%. The gap between full enforcement of the current restrictions and this coalition-wide export embargo reflects the products that the current lists leave untouched. It makes product scope a natural margin for the sanctions-design analysis that follows.

The next section turns from measurement to sanctions design. The model does not determine whether the geopolitical objective of weakening Russia is worth the associated income costs. It can, however, compare available instruments by how much additional loss they impose on Russia for a given cost to the coalition, and distinguish measures the coalition controls directly from those that require the cooperation of third countries.

7 Sanctions Design: Strengthening Export Bans

The enforcement gap quantifies how much transshipment dampens the impact of the current export bans, but it does not show which response is most effective. The coalition can tighten enforcement against current routes and hubs, seek sanctions alignment from important intermediary countries, or change the product list itself. These margins differ in what they change and in whether their implementation depends on action outside the coalition.

We begin with the current product lists and compare route-specific enforcement, country-wide hub shutdowns, and coalition enlargement. Since enlargement cannot be implemented by the existing coalition alone, we then use the EU–Türkiye Customs Union to compare the cost to Türkiye of joining while the CU remains in place with the cost of remaining outside after a CU-to-FTA downgrade. Finally, we turn to product scope, the margin most directly controlled by the sanctioning countries, and compare its effects and costs with those of policies that depend on third-country behavior.

7.1 Enforcing the Current Bans: Routes, Hubs, and Coalition Enlargement

The most direct response to transshipment is to target the routes that currently carry banned goods. Enforcement of this kind is already an important part of the sanctions regime. In October 2024, for example, the U.S. Treasury designated 275 individuals and entities across 17 jurisdictions for supplying Russia’s military-industrial base or moving sanctioned goods through intermediary networks.⁴² The EU’s Common High Priority Items list similarly supports exporters’ due diligence and the anti-circumvention work of customs authorities in partner countries.⁴³ Route-specific enforcement is naturally directed at what the authorities can observe. A route that carries no flow today is not part of the current rerouting network, but it may become attractive once existing routes are closed.

Our model does not contain individual firms, so we represent route-specific enforcement at the origin-product-hub level. We identify every route used in the initial observed-evasion equilibrium and make that particular combination unavailable. The affected flows then move to their cheapest remaining routes. This is deliberately a strong exercise. Every active route is identified and closed with certainty, so the result is an upper bound on what one static round of route-specific enforcement can achieve at this level of aggregation.

⁴² See “Treasury Takes Aim at Third-Country Sanctions Evaders and Russian Producers Supporting Russia’s Military Industrial Base,” U.S. Department of the Treasury, October 30, 2024.

⁴³ See the European Commission’s [List of Common High Priority Items](#).

TABLE 5. CLOSING HUBS AND ENLARGING THE COALITION

Country	Increase in Russia's Real Income Loss			
	Hub Shutdown (pp)	Further Effect of Joining (pp)	Joining in Total (pp)	Hub Shutdown (% of Total)
Türkiye	0.080	0.046	0.126	63.5
Belarus	0.027	0.092	0.119	22.6
China	0.031	0.650	0.681	4.6
United Arab Emirates	0.015	0.009	0.025	62.7
Serbia	0.003	0.010	0.013	23.5

Notes. This table compares the increase in Russia's real-income loss from shutting down a country's hub role with the total increase when that country joins the coalition, in both cases relative to observed evasion. "Hub shutdown" excludes the country from all indirect routes while leaving its own exports unchanged. "Further effect of joining" is the additional increase when the country also adopts the EU regime's export bans on its own exports to Russia and Belarus. It is computed as the difference between the joining and hub-shutdown counterfactuals and therefore contains all GE interactions between the two components. "Joining in total" is the difference between the joining counterfactual and observed evasion. The first three numeric columns are basis points of Russia's baseline real income. The last column reports hub shutdown as a percentage of the total joining effect, computed from unrounded values. The results are discussed in Section 7.

Russia's real-income loss rises from 1.26% to 1.38%. Even this comprehensive first round therefore recovers only 30% of the enforcement gap. The limited effect follows from endogenous rerouting, because once the best routes are shut down, previously unused routes become the cheapest alternatives, and flows move there instead of disappearing.

Closing a country as a hub reaches further. A hub shutdown makes the country ineligible as an intermediary for every banned origin-product flow, including routes that were initially inactive, while leaving its own exports to Russia unchanged. Coalition accession adds a second margin. In counterfactuals where a hub joins the coalition, the country adopts the EU regime's export bans on its own exports to Russia and Belarus and is no longer eligible as a hub for banned coalition-origin flows. Its newly banned origin goods may still be rerouted through the remaining non-coalition hubs.

Table 5 compares the effect of shutting down the hub role with the total effect of coalition accession for Türkiye, the United Arab Emirates, and China, the three largest hubs under observed evasion. The difference between the two counterfactuals is the additional effect of applying the EU export bans to the candidate's own goods. For Türkiye, the hub shutdown already delivers 63% of the total accession effect, and for the United Arab Emirates it delivers 63%. Thus, most of the effect in these two countries comes from ending their role as intermediaries. China displays the opposite pattern. Its hub shutdown accounts for only 5% of the total accession effect because restricting Chinese-origin exports to Russia matters much more than removing China's intermediation role.

Note that route and hub closures change the enforcement of the bans already in place and therefore cannot impose a larger loss than the no-transshipment loss under the current

product lists. Coalition enlargement is not subject to this ceiling because it also adds new sanctioning origins. If Türkiye, the United Arab Emirates, and China all adopted the EU regime, Russia’s real-income loss would rise to , above the 1.66% no-transshipment loss under the current coalition’s restrictions.

However, coalition enlargement cannot be implemented by the existing members alone, because it depends on the cost borne by the candidate and on the economic leverage available to the coalition. We put these two magnitudes into perspective for Türkiye, the largest hub under observed transshipment.

7.2 The EU–Türkiye Customs Union as a Policy Lever

Türkiye is relevant because it carries the largest share of rerouted flows, earns the largest intermediation rents under observed transshipment, and has preferential access to the European market through the EU–Türkiye Customs Union (CU). In what follows, we do not model negotiations over accession. Instead, we use the model to compare two policy outcomes. In the first, Türkiye adopts the EU sanctions regime, gives up its hub role, and retains the CU. In the second, it remains outside the coalition and retains its hub role, but the CU is replaced by an ordinary free trade agreement. This exercise compares the economic value of the CU with the cost Türkiye bears from sanctions alignment.

Under observed transshipment, Türkiye’s real income is 0.01% above the no-sanctions baseline, largely because of the rents earned on rerouted trade. Relative to remaining outside the coalition under the current CU, adopting the EU export bans and giving up its hub role would cost Türkiye 0.06% of real income.

To quantify the second outcome, we replace the CU with an ordinary free trade agreement. Tariffs remain close to zero on originating goods, but preferential rules of origin begin to apply. Following the European Commission, we represent this change by a 2.0% increase in bilateral industrial-goods trade costs and apply it in both directions.⁴⁴ Because this counterfactual leaves services, investment, and the remainder of the bilateral relationship unchanged, it captures only part of the cost of losing CU access. Even so, refusing to join the coalition and accepting the CU-to-FTA downgrade lowers Türkiye’s real income by 0.38% relative to the status quo, about 6.7 times the cost of joining.

Under this conservative specification, joining while retaining the CU is therefore substantially less costly to Türkiye than remaining outside after the downgrade. The comparison does not predict the outcome of negotiations between the EU and Türkiye. It shows that the economic value of the CU is large relative to the modeled cost of sanctions alignment.

⁴⁴ The Commission reviews estimates of the cost of introducing rules of origin, ranging from roughly 2% to 6%, and uses 2.0% as the conservative value for replacing the CU with a free trade agreement (European Commission, 2016, p. 28). Cadot and de Melo (2008) report tariff equivalents of 3% to 5%. The broader trade effects of the CU are larger still (see, e.g., Larch et al., 2021).

A CU downgrade would also be costly to the EU, which we quantify after considering the product-scope margin.

7.3 Widening the Scope of the Export Bans

The two policies considered so far require conduct outside the existing sanctioning group to change. Active routes must shut down, or a candidate country must adopt the restrictions. By contrast, the current sanctioning group can widen its product list without changing the behavior of non-coalition hubs. We therefore hold the calibrated transshipment technology fixed as the list expands. Transshipment remains available, and the exercise isolates product scope under observed transshipment.

We evaluate product-scope expansions against two benchmarks. The first is the 1.38% loss obtained after closing every route used in the initial observed-transshipment equilibrium, our upper bound for one round of enforcement against all initially active routes. The second is the 1.66% no-transshipment loss under the coalition’s current product restrictions. Reaching the latter under observed transshipment would restore the loss that a standard no-transshipment quantification assigns to the current restrictions.

We conduct the exercise for the EU and the countries that mirror its sanctions, because this group applies a common product list and our sanctions data distinguish partial from full bans for every covered product. We assume that the mirroring countries continue to align with additions to the EU list. The candidate set consists of the 3,193 products that the group does not fully ban by the end of 2024. When product k is added, every origin in the group, denoted by $\mathcal{B}^k$, prohibits its exports of that product to Russia. The restrictions imposed by the remaining coalition countries are left unchanged.

Ranking products. Candidate products differ in their importance for Russian expenditure, Russia’s dependence on the sanctioning origins, their trade elasticities, and the routes available for rerouting around a new ban. We rank them by the (first-order) reduction in Russian real income per dollar of the group’s baseline exports newly covered by the ban. The score underlying the ranking is a partial-equilibrium rule used only to order products, and its denominator measures the amount of trade brought under the ban rather than the real-income cost to the sanctioning group. We evaluate every reported expansion in the exact general equilibrium below.

Russia’s real income is $W_R = X_R/P_R$. Holding wages and nominal expenditure at their baseline values, the nested demand system implies, around the baseline,

$$\Delta \ln W_R = -\Delta \ln P_R \approx -\sum_k s_R^k \Delta \ln P_R^k,$$

where $s_R^k = X_R^k/X_R$ is product k 's baseline share in Russian expenditure. A ban on product k alone therefore lowers Russian real income, to a first order, by $s_R^k \Delta \ln P_R^k$.

To calculate the product-price effect, $\Delta \ln P_R^k$, let $b^k = \sum_{e \in \mathcal{B}^k} \pi_{eR}^k$ denote the sanctioning origins' baseline share in Russia's expenditure on product k . Setting $\hat{w}_e = 1$ in equation (21), a ban with no transshipment (NT) then raises the product-level price index by

$$\Delta \ln P_R^{k,NT} = -\frac{1}{\theta^k} \ln(1 - b^k). \quad (31)$$

Under observed transshipment (OT), the newly banned origins can use the calibrated transshipment technology. The corresponding fixed-wage price change is

$$\Delta \ln P_R^{k,OT} = -\frac{1}{\theta^k} \ln \left[1 - b^k + \sum_{e \in \mathcal{B}^k} \pi_{eR}^k \left(\frac{\tau_{eR}^{k'}}{\tau_{eR}^k} \right)^{-\theta^k} \right], \quad (32)$$

where $\tau_{eR}^{k'}$ is the final importer-facing cost of the selected route from equation (17), including its markup. The additional sum is the contribution of the newly banned origins to the post-ban price-index denominator after rerouting. It is zero under no transshipment and is precisely the term omitted by an evaluation that neglects transshipment. For newly covered cells, we floor the final cost at the cell's pre-ban direct cost, so that adding a product to the list can never reduce its delivered cost. For each candidate product, we recompute the selected intermediary and its markup under the new ban, holding the calibrated hub wedges and β fixed.

Under scenario z , the first-order real-income damage is $s_R^k \Delta \ln P_R^{k,z}$. We compare this damage with v^k , the baseline value of the group's exports to Russia that the ban would newly restrict, expressed as a share of the bloc's total absorption. Since the denominator used to express this exposure is common across products, the resulting order is the same as when exposure is measured in dollars. The score is

$$r_z^k = \frac{s_R^k \Delta \ln P_R^{k,z}}{v^k}, \quad z \in \{NT, OT\}. \quad (33)$$

The transshipment-adjusted list ranks products by r_{OT}^k . Instead, ranking by r_{NT}^k yields the list that neglects transshipment, following the standard no-transshipment counterfactual logic. The scores determine only the order in which products enter. For every reported expansion, we impose the corresponding additional bans and solve the exact general equilibrium.

Results. Figure 11 traces the transshipment-adjusted expansion under observed transshipment. The expansion path is steep at first and then flattens, showing that the

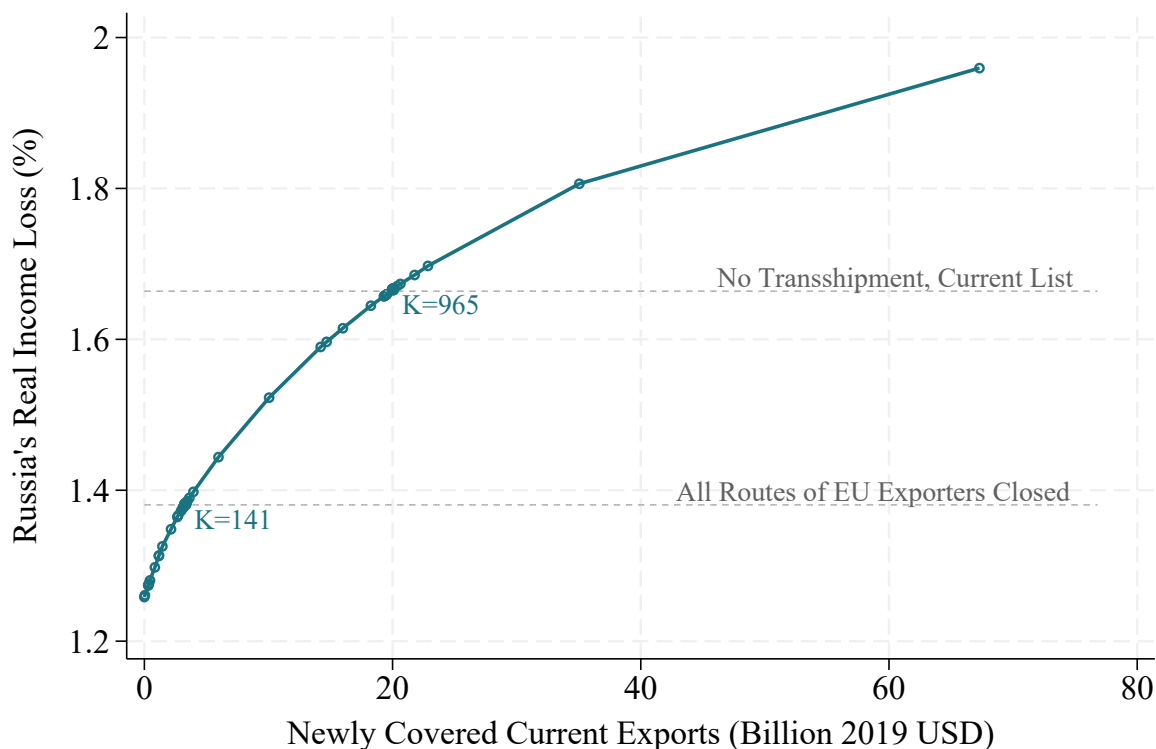


FIGURE 11. WIDENING THE PRODUCT SCOPE UNDER OBSERVED TRANSSHIPMENT

Note: This figure reports Russia's real-income loss as the EU plus mirroring group widens the product scope of its export bans. The series ranks candidate products by transshipment-adjusted damage per unit of exposure and evaluates each reported expansion under observed transshipment. Each point is an exact general-equilibrium solution in which the bloc bans the first K products of the list, where K is the number of products added to the current EU list. The series begins with the current restrictions under observed transshipment. The lower horizontal line marks the loss after closing every route used in the initial observed-transshipment equilibrium. The upper horizontal line marks the no-transshipment loss under the coalition's current product restrictions. The labeled markers identify the expansions that reach these benchmarks, at $K = 141$ and $K = 965$. The horizontal axis reports the cumulative baseline value of the newly covered exports in billions of 2019 USD. The vertical axis reports Russia's real-income loss in percent of baseline real income.

additional effect on Russia per unit of the group's export exposure is concentrated among the first products added.

Under observed transshipment, the top 141 products alone in the transshipment-adjusted ranking, covering USD 3.3 billion of baseline exports, suffice to raise Russia's loss to the level achieved by closing every initially used route. This increases the baseline export value covered by the bloc's current list by 4%. The top 965 products, covering USD 20.0 billion, restore the no-transshipment loss under the current restrictions. This larger package increases the baseline export value covered by the list by 25%.

The partial-equilibrium (fixed-wage) scores follow the exact effects closely. As a direct check, we impose selected candidate bans one at a time on top of the current restrictions and solve the full general equilibrium. The ranking of their exact damage per dollar has

TABLE 6. COMPARING WAYS TO STRENGTHEN THE EXPORT BANS

Instrument	Add. Russian Loss (% of Transship. Gain)	Add. Median EU Member Loss (% of Real Income)
Close initial routes	+0.122pp (30%)	0.0050%
Türkiye joins, including hub removal	+0.126pp (31%)	0.0024%
Transshipment-adjusted product expansion	+0.123pp (30%)	0.0016%

Notes. This table compares the policy exercises considered in this section. The first column reports the additional loss of Russian real income relative to observed transshipment, in percentage points, with the share of the transshipment gain this recovers in parentheses. The second column reports the additional real-income loss of the median member state relative to observed transshipment, in percent of its baseline real income. In 2019 dollars, the total EU-27 real-income losses are USD 0.21 billion for the product expansion and USD 0.28 billion for Türkiye’s accession. Costs to other coalition members and administrative enforcement costs are not included. The final column asks whether implementation requires non-coalition conduct to differ from the observed-transshipment equilibrium. Closing all initially used routes requires the active third-country hubs to shut down those routes, while Türkiye’s accession requires it to adopt and implement the EU restrictions. The product-scope expansion is marked “No” because non-coalition hubs remain available and transshipment continues under the same calibrated technology. Each row is an exact general-equilibrium solution. The results are discussed in the text.

a rank correlation of 0.88 with the fixed-wage score across the 18 solved products with positive exposure. The score also tracks the incremental damage along the expansion in Figure 11, with a rank correlation of 0.992.⁴⁵

To isolate the contribution of product ordering, we evaluate the transshipment-adjusted list and the list that neglects transshipment, both under observed transshipment, at the exposure budget of the 141-product package (the smallest transshipment-adjusted expansion that reaches the route-closure benchmark). The largest package neglecting transshipment that fits this budget contains 66 products and produces 22% less additional damage. A product can rank highly on the no-transshipment score even when the available routes absorb much of its intended price effect. The two lists converge as the scope widens and coincide by construction once every candidate product is covered. Ordering therefore matters over the steep initial range, not at budgets approaching full coverage.

7.4 Comparing the Instruments

The three exercises generate almost the same additional loss for Russia, which makes their modeled costs to the EU directly comparable. Importantly, they differ in whether implementation requires non-coalition conduct to change. Table 6 reports the relevant losses.

⁴⁵ For the single-product check, we solve the first eight products of the transshipment-adjusted list one by one and then products at roughly doubling ranks, down to two near the bottom of the list. Damage per dollar is undefined for candidates with zero baseline exposure.

The transshipment-adjusted product package is chosen to reach the route-closure benchmark, while Türkiye's accession produces almost the same additional loss for Russia. Among the three exercises, the product expansion has the lowest computed real-income cost for the EU. This ordering is consistent with the logic of the product ranking. Route enforcement disrupts every rerouting route used in the initial equilibrium, whereas the product package concentrates new restrictions on goods that impose high damage on Russia relative to the coalition exports they bring under the ban.

The Customs Union calculation provides a separate perspective on the economic leverage available over Türkiye. Replacing the CU with a free trade agreement would cost the EU at least USD 1.61 billion and would leave Russia's loss essentially unchanged. For Türkiye, the same downgrade is substantially more costly than joining the coalition while retaining the CU. The CU therefore matters quantitatively for Türkiye's incentives, but using this policy lever would also be costly to the EU. The product expansion reaches a comparable loss for Russia at a much lower EU cost and without requiring Türkiye's behavior to change.

Closing the routes used in the initial equilibrium leaves most of the enforcement gap open because flows move to unused alternatives. Coalition enlargement can go further, especially when the candidate is an important supplier as well as a hub, but it depends on the candidate adopting and implementing the restrictions. A targeted expansion of the product list can produce a similar additional loss while non-coalition hubs continue to operate under the observed transshipment technology. Among the three exercises considered here, the scope expansion reaches the benchmark loss at the lowest computed cost to the EU. However, whether any of these costs are worth bearing depends on the geopolitical objective and lies outside the scope of our analysis.

8 Conclusion

This paper quantifies how third-country transshipment changes the effectiveness and design of export bans. We combine Russian transaction-level customs records with a new product-level dataset on the legal scope and timing of EU export bans. Because the customs records report both the origin and dispatch country of each shipment, we can observe an important form of transshipment directly. At least 26% of the pre-war monthly value of fully banned EU-origin products continues to reach Russia after the restrictions take effect. transshipment initially relies heavily on in-transit transshipment, but shifts over time toward re-exporting through a concentrated set of third-country hubs.

We estimate that the export bans cut the affected flows by roughly 80 percent relative to the comparison groups, and we use this response together with the observed routes to discipline an HS6-level general-equilibrium model with endogenous transshipment. The calibrated model implies that the bans raised median effective trade costs from the EU to Russia by about 69%. The model distinguishes legal substitution toward other suppliers and products from evasion through intermediaries, and allows hubs to earn rents from facilitating rerouted trade.

The quantitative results reveal a sizable enforcement gap. Under observed evasion, Russia’s real income falls by 1.26% relative to a world without the sanctions. With no transshipment, the same product restrictions raise the loss to 1.66%, an increase of 32%. The decline in targeted trade is much larger than the decline in the goods available to Russia. Imports of banned coalition-origin products fall by 76%, but substitution toward other suppliers, domestic production, and other products limits the fall in total imports and in access to military-relevant goods. Under our benchmark markup assumption, intermediary hubs earn up to USD billion from rerouted trade.

The policy exercises show why enforcement directed at the current network has limited returns. Even if every route used under observed evasion is identified and closed, only 30% of the enforcement gap is recovered because trade moves to previously unused routes. Closing an entire hub reaches further, while bringing a country into the coalition also restricts its own exports. The importance of these two channels differs across countries. Türkiye and the United Arab Emirates matter mainly as intermediaries, while China’s own exports account for most of the effect of its coalition accession.

Changing the product list provides a policy margin controlled directly by the existing sanctioning group. A transshipment-adjusted expansion of 141 products generates approximately the same additional loss for Russia as closing every initially used route and does so at a lower modeled cost to the EU. A broader expansion of 965 products restores, under continued transshipment, the no-transshipment loss under the current restrictions.

At a common exposure budget, a product ranking that ignores transshipment generates 22% less additional damage.

The broader lesson is that export bans act on an adjusting trade network. Closing one route changes the alternatives available to firms, and the rents from transshipment affect the incentives of intermediary countries. Evaluating sanctions solely from the legal restrictions or the trade routes observed today can therefore misstate both their aggregate effects and the relative merits of alternative policy responses. Our empirical measure captures evasion for which the original country of origin remains visible, and our counterfactuals are static rather than a model of repeated enforcement and network formation. The same approach can be applied more broadly when tariffs, sanctions, or other discriminatory trade barriers can be circumvented through third countries.

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A Further Details on the Data

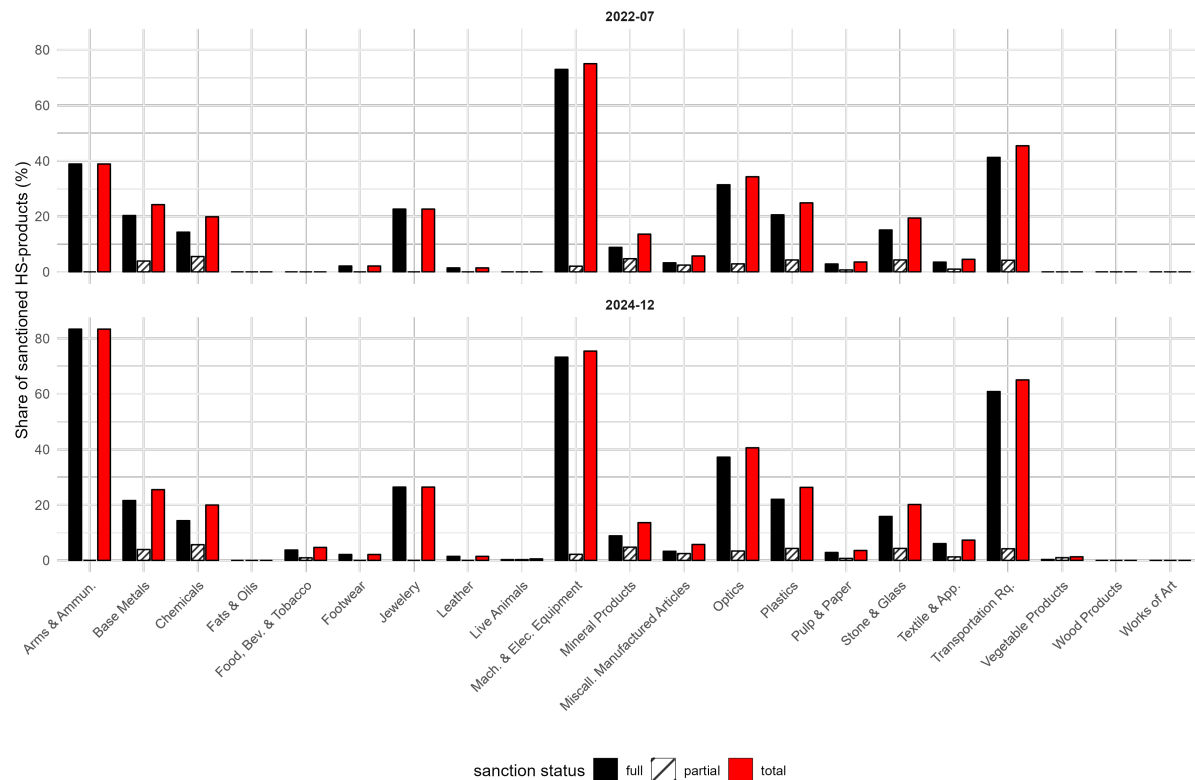


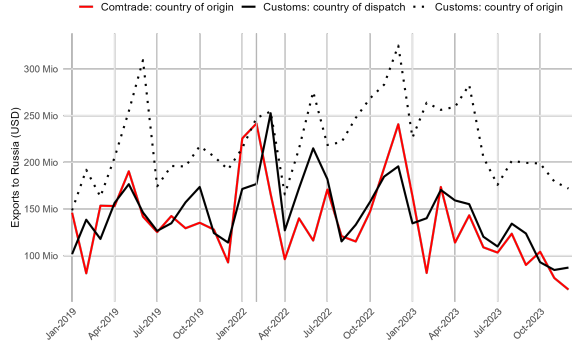
FIGURE A.1. SECTORAL COVERAGE OF EU-BLR SANCTIONS BY STATUS AND EIF DATE

Note: This figure reports, by sector, the share of EU-sanctioned products vis-a-vis Belarus at the 6-digit HS2017 level, distinguished by sanction status. Black bars indicate the share of fully sanctioned products, striped bars the share of partially sanctioned products, and red bars the total share of sanctioned products. The upper panel includes all sanctions that came into force by July 2022, while the lower panel includes all sanctions in effect by December 2024.

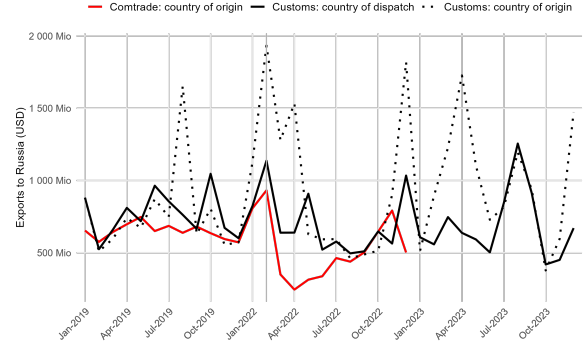
1st package	Feb, 23 th 2022	-
2nd package	Feb, 25 th 2022	dual-use/high-tech*, oil technologies*, aviation*
3rd package	March, 1 st 2022	banknotes*
compliance	March, 9 th 2022	maritime navigation*
4th package	March, 15 th 2022	steel and iron* †, luxury*
5th package	April, 8 th 2022	goods strength. the ind. capability*, aviation*, luxury*, oil technologies*, dual-use/high-tech*, coal†, strategic†, steel & iron†
6th package	June, 3 rd 2022	dual-use/high-tech*, petroleum oils†, strategic†
maintenance & alignment	July, 21 st 2022	dual-use/high-tech*, goods strengthening the ind. capability*, oil technologies*, gold†
8th package	Oct, 10 th 2022	aviation*, dual-use/high-tech*, firearms*, goods strengthening the ind. capability*, iron and steel†, strategic†
9th package	Dec, 16 th 2022	dual-use/high-tech*, aviation*, goods strengthening the ind. capability*, petroleum oils†, iron and steel†
10th package	Feb, 25 th 2023	dual-use/high-tech*, aviation*, goods strengthening the ind. capability*, strategic†
11th package	June, 23 rd 2023	dual-use/high-tech*, firearms*, goods strengthening the ind. capability*, luxury*, strategic†, steel and iron†
12th package	Dec, 18 th 2023	dual-use/high-tech*, goods strengthening the ind. capability*, strategic†, diamonds†
13th package	Feb, 23 rd 2024	dual-use/high-tech*, goods strengthening the ind. capability*
14th package	June, 24 th 2024	dual-use/high-tech*, goods strengthening the ind. capability*, strategic†

TABLE A.1. TIMELINE OF EU SANCTIONS PACKAGES AGAINST RUSSIA.

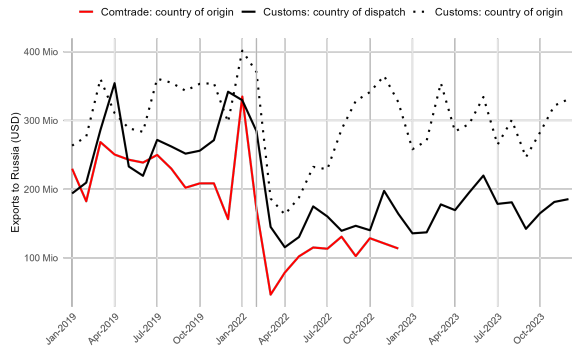
Note: * export sanction, † import sanction



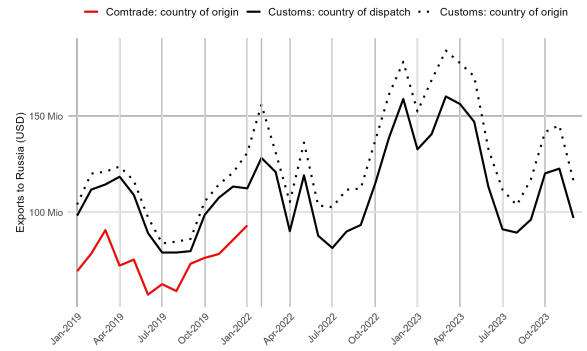
(A) BRAZIL



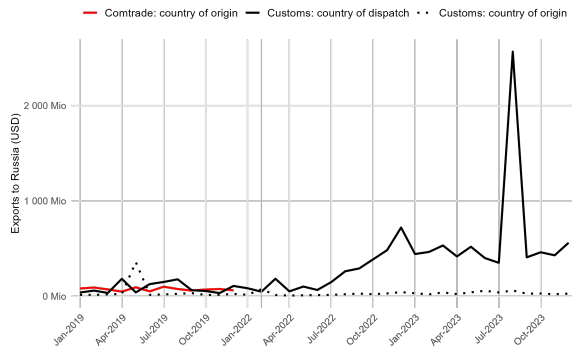
(B) SOUTH KOREA



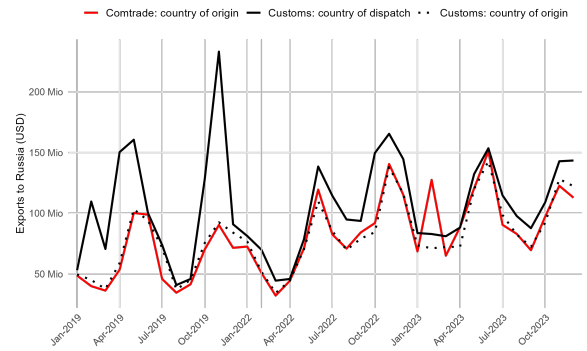
(C) VIETNAM



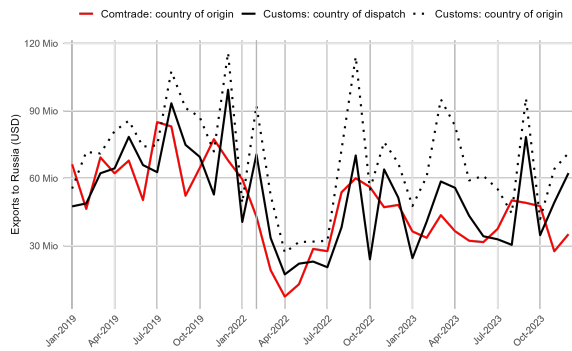
(D) ECUADOR



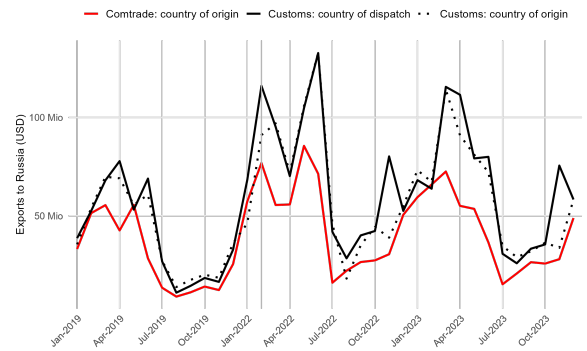
(E) UNITED ARAB EMIRATES



(F) AZERBAIJAN



(G) CHILE



(H) EGYPT

Note: The figure shows Russian imports from the customs data (by month, year, and country of dispatch & origin) alongside exports to Russia from the same countries reported in UN Comtrade. Data are shown for 2019 (pre-period) and 2022–2023 (post-onset of the Russian war) for other non-EU, non-coalition exporters with available Comtrade data. Trade values are in millions of US dollars.

FIGURE A.2. VALIDATION OF RUSSIAN CUSTOMS DATA (OTHER EXAMPLES)

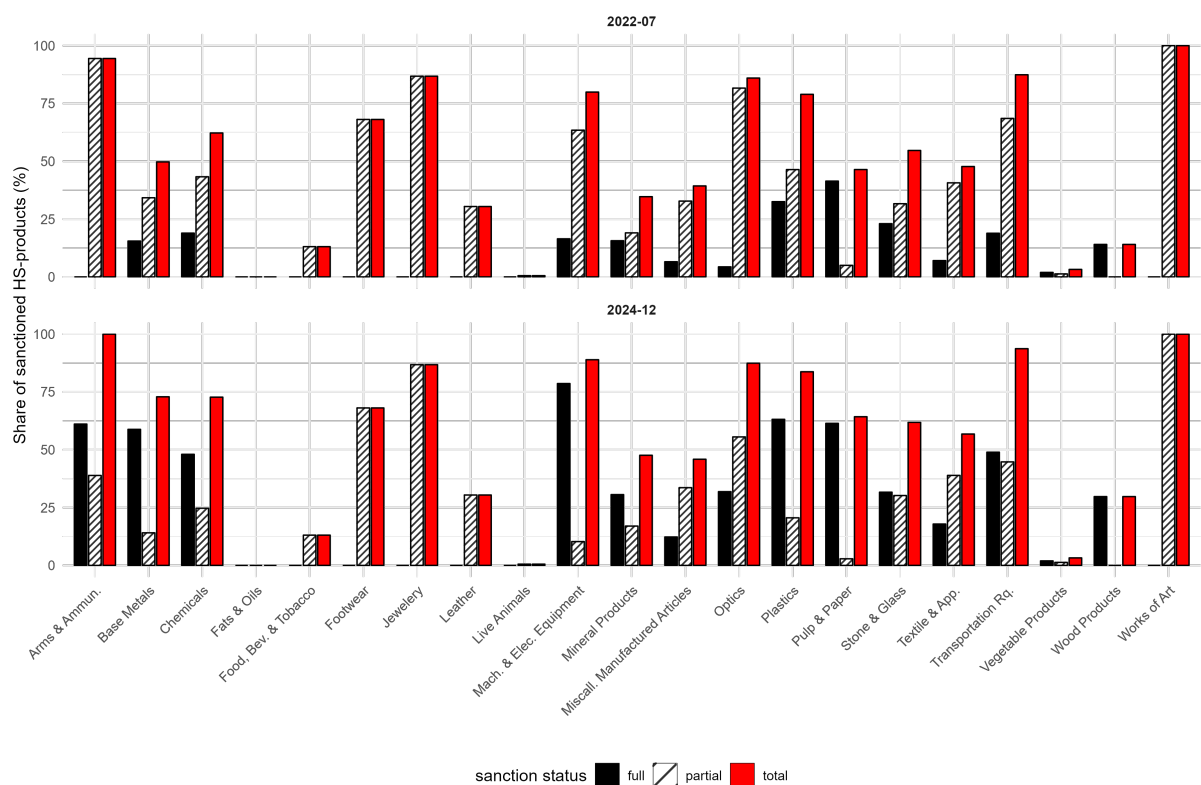


FIGURE A.3. EU EXPORT BANS BY SECTORS

Note: The figure reports, by sector, the share of HS6 products covered by EU export bans. Black bars indicate fully banned products, striped bars partially banned products, and red bars total banned products. The upper panel shows coverage as of July 2022, and the lower panel as of December 2024.

B Additional Figures

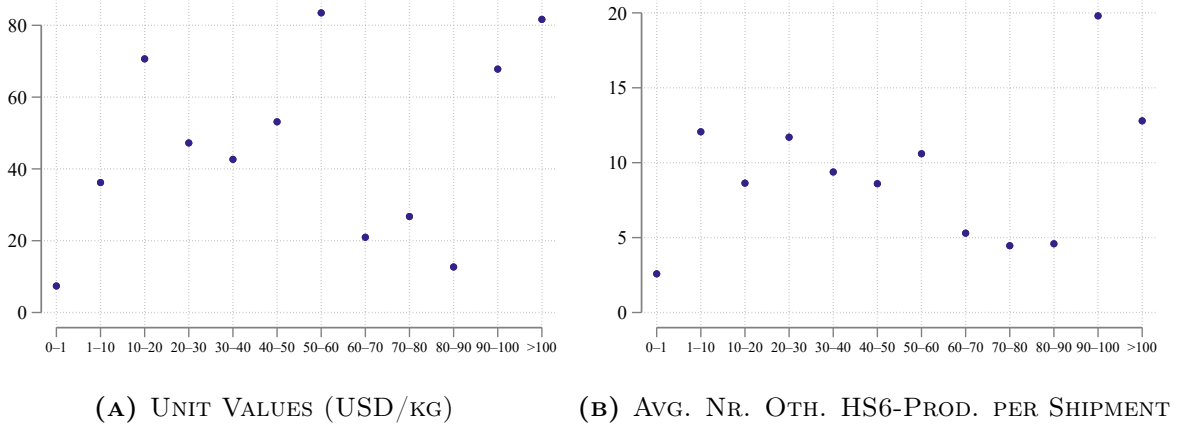


FIGURE B.1. EVASION AND PRE-WAR CHARACTERISTICS OF EXPORT-BANNED PRODUCTS

Note: Bins are constructed over an evasion measure defined for each HS6 product as the ratio of its average monthly Russian imports of EU-origin products in the post-export-ban period (August 2022–December 2024) to the corresponding average in 2019. The sample consists of the 668 HS6 products fully subject to EU export bans by July 2022. Each point reports the unweighted mean of the characteristic shown on the y-axis within a bin of the evasion measure, except for unit values, which are reported as medians within bins. All product characteristics refer to 2019.

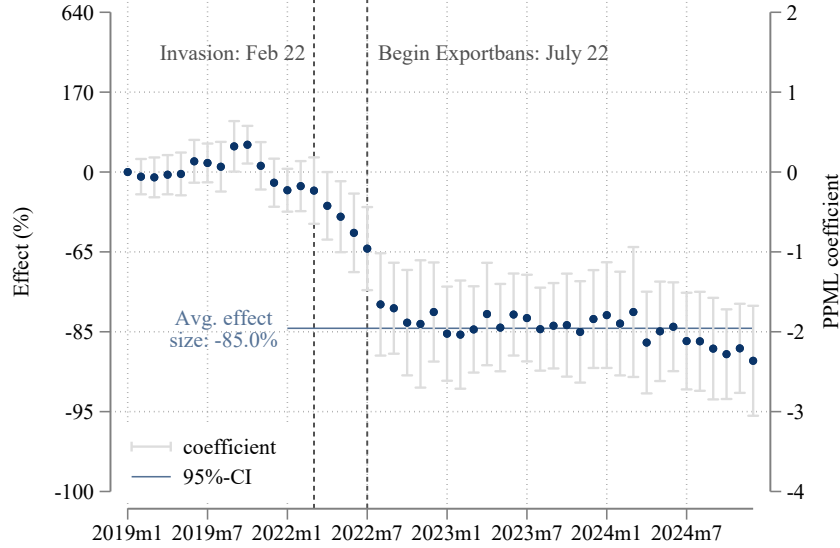
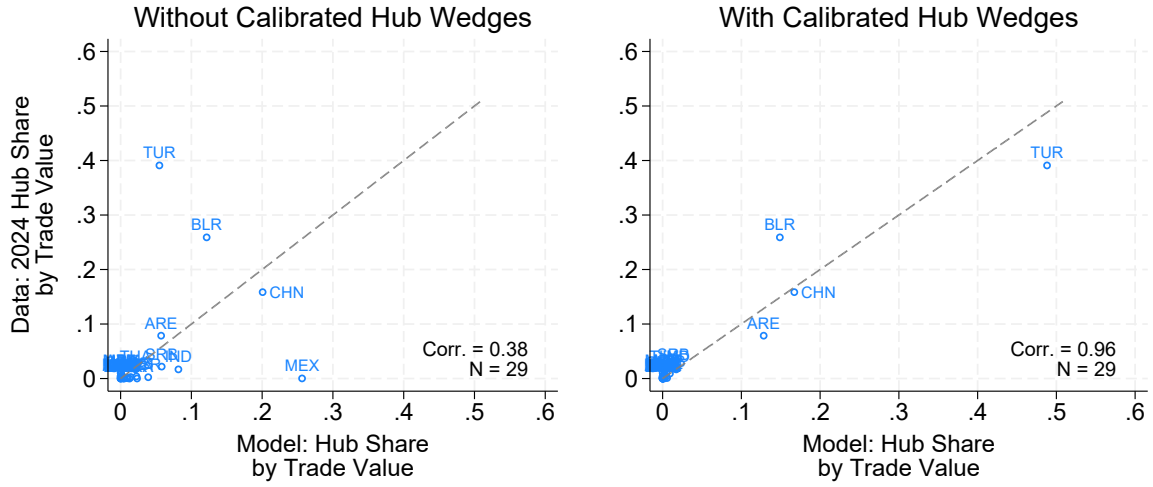
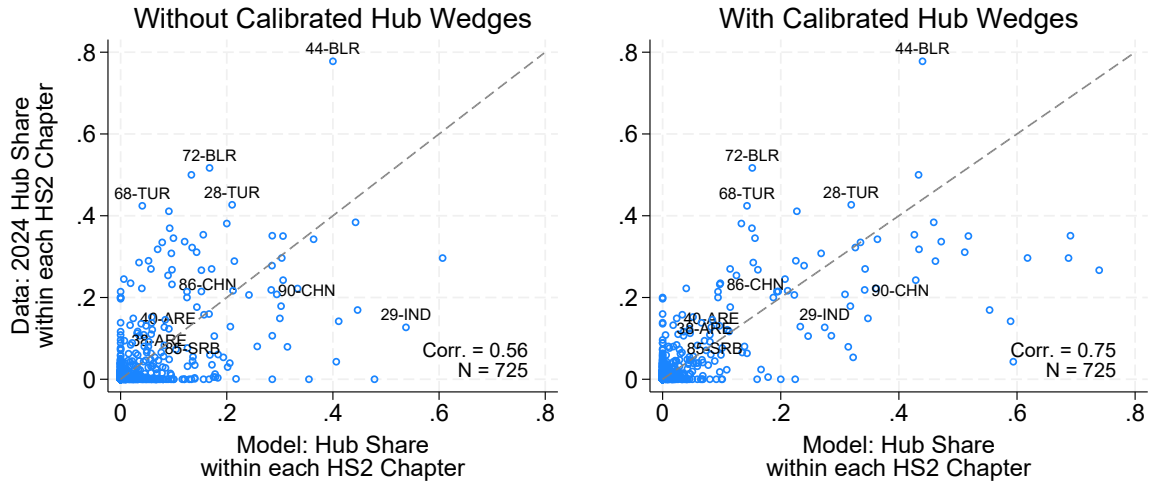


FIGURE B.2. EVENT-STUDY ESTIMATION RESULTS (QUANTITIES)

Note: The figure shows the estimated $\hat{\beta}_m$ coefficients of the event-study specification (2), but using export quantities rather than values as the dependent variable. The coefficient for the first month (January 2019) is normalized to zero. The raw coefficients are shown on the right y-axis. The left y-axis reports the corresponding partial-equilibrium trade-flow changes, $(e^{\hat{\beta}_m} - 1) \times 100\%$.



(A) HUB SHARES BY TRADE VALUE, ALL COALITION ORIGINS



(B) HUB SHARES WITHIN EACH HS2 CHAPTER, BY CHAPTER-HUB PAIR

FIGURE B.3. UNTARGETED FIT OF THE CALIBRATED HUB WEDGES: VALUES AND CHAPTERS

Note: This figure repeats the comparison of Figure 10 for two further cuts of the same data, in each panel without the calibrated hub wedges (left) and with them (right). Panel B.3a aggregates by hub over all coalition origins and weights by trade value. The data values are observed 2024 dollar flows, and the model values are each specification’s own counterfactual gross flows to Russia (which include the hub markups). Panel B.3b disaggregates by HS2 chapter and hub over all coalition origins, counts routing relations as in Figure 10, and shows the 25 chapters with at least 25 matched cells. Shares sum to one within each chapter, and each chapter is given by its two-digit HS code. For legibility, labels mark ten high-share chapter-hub pairs, at most two per hub. Sample restrictions and the country sample are as in Figure 10. The dashed line is a 45°-line. “Corr.” reports the correlation value, an unweighted Pearson correlation across the plotted points, and “N” the number of plotted points.

C Additional Tables

TABLE C.1. TRANSSHIPMENT HUBS: RENTS, EXPORTS, AND REAL INCOME

Hub	Use Share	Rents USD M	Export Gain USD M	With Hub Rents: Real Income Change (%)			Main
				Total	Real Wages	Rents	Total
TUR	30.50%	1,028.5	1,799.6	0.189%	0.021%	0.168%	0.012%
BLR	19.56%	332.2	2,309.7	-0.570%	-0.979%	0.409%	-0.976%
CHN	13.21%	265.3	16,345.5	0.006%	0.005%	0.001%	0.005%
ARE	7.82%	186.2	424.6	0.090%	0.039%	0.051%	0.038%
SRB	6.24%	41.5	485.9	0.159%	0.073%	0.086%	0.077%
IND	5.56%	58.2	1,617.5	0.006%	0.003%	0.002%	0.003%
MDV	0.00%	0.0	0.0	0.007%	0.007%	0.000%	0.007%
UZB	1.00%	8.0	333.3	0.018%	0.004%	0.013%	0.004%
THA	1.96%	9.0	788.8	0.009%	0.008%	0.001%	0.008%
AZE	0.29%	0.4	90.6	0.048%	0.047%	0.001%	0.046%

Notes. This table reports the ten largest transshipment hubs in the 2024 customs data, ranked by their observed hub-use shares, with the model's calibrated values reported for each. Hub-use shares give each hub's share of the rerouted banned flows of the EU plus mirroring group, with each positive origin-product flow counted once. Hub rents are the intermediary premium income in the observed-evasion equilibrium, in millions of model-2019 USD. Export gains are the change in each hub's own exports to Russia relative to the baseline, summed across all products. These goods originate in the hub itself, whereas rerouted coalition goods keep their coalition origin and are not included in this column. The last three columns split the change in real income, $(\hat{W} - 1) \times 100$, into the direct contribution of the rents and the change in real wages. The split is exact within the model, as baseline rents are zero. The real-wage component still reflects the equilibrium effects of spending the rents, so it is not based on a rent-free counterfactual. But recalibrating the model without hub rents ($\beta = 0$) yields gains close to the real-wage component for most hubs shown. Components are computed from unrounded values. Belarus is itself sanctioned throughout, which explains its real income loss. The results are discussed in Section 6.

TABLE C.2. ROBUSTNESS OF THE MAIN RESULTS

Specification	Observed-Transshipment	No-Transshipment	No vs. Observed	Banned-Import
	Loss	Loss	Transshipment	Decline
Main specification	1.26%	1.66%	32%	76%
$\nu = 0.5$	1.26%	1.67%	32%	76%
$\nu = 0.9$	1.26%	1.66%	32%	76%
$\nu = 2.2$	1.25%	1.66%	32%	76%
$\sigma_M = 1.2$	1.26%	1.67%	32%	76%
Home-share floor 5%	1.24%	1.70%	37%	75%
No hub markups ($\beta = 0$, recalibrated)				
Wartime demand ($\hat{\alpha}_R^M = 3$)	1.32%	1.72%	31%	78%
Trade elasticities halved	2.29%	2.92%	28%	78%

Notes. This table reports the robustness of the quantification in Table 3 to alternative parameter choices. Except for the wartime-demand exercise, each specification refits the relative hub wedges to the observed 2024 routing and their common scale to the triple-difference estimate, as in Section 5.3. The wartime-demand exercise instead holds the benchmark transshipment technology fixed in order to isolate the effect of the demand shift. The main specification sets $\nu = 1.1$, $\sigma_g = 2.2$ for all groups, a 10% floor on home shares, and $\beta = 1$. “No hub markups” recalibrates the $\beta = 0$ version of the model, in which intermediaries earn no rents by construction. “Wartime demand” triples Russia’s demand shifter for the military-relevant group, holding the calibrated evasion technology fixed. Its losses are measured against the recomputed shifted economy’s own no-sanctions baseline equilibrium. “Trade elasticities halved” scales the elasticities θ^k that govern the counterfactual responses by one half, leaving the calibrated demand system unchanged. In this version, the trade elasticities are floored at $(\sigma_{g(k)} - 1) + 0.1$, which preserves the condition $\theta^k > \sigma_{g(k)} - 1$ required by the product-expenditure recovery. The floor binds for 227 products. The first two columns report Russia’s real income loss in percent relative to the baseline without the sanctions, under observed evasion and under full enforcement. The third column reports the increase in the loss when moving from observed evasion to full enforcement, as a percentage of the observed-evasion loss. The last column reports the decline in Russia’s imports of banned products from coalition countries under observed evasion. The results are discussed in Section 6.

TABLE C.3. ROBUSTNESS OF GROUP-LEVEL REAL ABSORPTION

Specification	Decline in Real Absorption (%)					
	Observed Transshipment			No Transshipment		
	M	C	O	M	C	O
Main specification	3.65%	1.95%	0.89%	4.04%	2.69%	1.14%
$\nu = 0.5$	2.35%	1.58%	1.09%	2.76%	2.13%	1.43%
$\nu = 0.9$	3.22%	1.83%	0.95%	3.62%	2.51%	1.24%
$\nu = 2.2$	5.97%	2.64%	0.51%	6.36%	3.71%	0.61%
$\sigma_M = 1.2$	3.72%	1.95%	0.89%	4.16%	2.69%	1.14%
Home-share floor 5%	3.60%	1.90%	0.89%	4.12%	2.75%	1.17%
Hub shares from re-exports	3.67%	1.98%	0.90%	4.04%	2.69%	1.15%
Wartime demand ($\hat{\alpha}_R^M = 3$)	3.62%	1.93%	0.87%	4.02%	2.67%	1.13%
Trade elasticities halved	6.60%	3.47%	1.65%	7.66%	4.64%	2.02%

Notes. This table reports the decline in Russia’s real absorption of the three product-group composites, $(1 - \hat{Q}_R^g) \times 100$, under observed evasion and under full enforcement, for every specification of Table C.2. The specifications, including the recalibration of the hub wedges to the same empirical moments in every arm, are described in the notes to that table. Group-level real absorption is defined in Section 5.2.3. The results are discussed in Section 6.

TABLE C.4. COUNTERFACTUAL SCENARIOS: EU MEMBER-STATE EXPORTS OF BANNED PRODUCTS TO RUSSIA AND REAL INCOME

Country	Full baseline USD M	Exports to Russia (%)						Real Income (%)		
		Full		Partial		Total				NT-OT (pp)
		OT	NT	OT	NT	OT	NT	OT	NT	
EU27	58,838	-76%	-100%	-27%	-23%	-39%	-48%	-0.036%	-0.048%	-0.009
AUT	1,656	-79%	-100%	-35%	-28%	-38%	-44%	-0.029%	-0.032%	-0.003
BEL	2,140	-74%	-100%	-40%	-37%	-37%	-47%	-0.019%	-0.025%	-0.005
BGR	298	-76%	-100%	-33%	-32%	-26%	-32%	-0.041%	-0.051%	-0.010
CYP	39	-92%	-100%	-39%	-45%	-37%	-41%	-0.024%	-0.026%	-0.002
CZE	2,930	-74%	-100%	-30%	-28%	-48%	-61%	-0.092%	-0.115%	-0.023
DEU	16,385	-73%	-100%	-24%	-19%	-40%	-51%	-0.043%	-0.054%	-0.011
DNK	583	-78%	-100%	-21%	-18%	-30%	-37%	-0.017%	-0.021%	-0.004
ESP	1,718	-72%	-100%	-28%	-27%	-33%	-41%	-0.014%	-0.018%	-0.004
EST	1,149	-88%	-100%	-31%	-29%	-52%	-57%	-0.326%	-0.361%	-0.035
FIN	3,470	-80%	-100%	-30%	-26%	-57%	-69%	-0.144%	-0.175%	-0.031
FRA	5,768	-80%	-100%	-27%	-24%	-43%	-52%	-0.023%	-0.028%	-0.004
GRC	227	-61%	-100%	-22%	-22%	-25%	-39%	-0.019%	-0.026%	-0.007
HRV	78	-80%	-100%	-30%	-28%	-11%	-13%	-0.009%	-0.008%	0.001
HUN	1,167	-75%	-100%	-29%	-27%	-35%	-44%	-0.048%	-0.062%	-0.013
IRL	142	-82%	-100%	-35%	-32%	-17%	-17%	-0.003%	-0.003%	0.000
ITA	5,226	-72%	-100%	-26%	-25%	-36%	-46%	-0.036%	-0.045%	-0.009
LTU	2,420	-83%	-100%	-24%	-21%	-40%	-46%	-0.458%	-0.514%	-0.056
LUX	237	-73%	-100%	-30%	-31%	-46%	-63%	-0.027%	-0.036%	-0.009
LVA	634	-83%	-100%	-17%	-15%	-30%	-35%	-0.178%	-0.209%	-0.031
MLT	25	-94%	-100%	-60%	-60%	-22%	-23%	-0.119%	-0.146%	-0.028
NLD	4,294	-77%	-100%	-32%	-30%	-36%	-45%	-0.069%	-0.085%	-0.016
POL	4,265	-74%	-100%	-29%	-28%	-37%	-47%	-0.080%	-0.098%	-0.018
PRT	245	-81%	-100%	-33%	-34%	-36%	-42%	-0.008%	-0.009%	-0.001
ROU	783	-64%	-100%	-27%	-27%	-33%	-45%	-0.036%	-0.048%	-0.012
SVK	679	-68%	-100%	-17%	-6%	-26%	-26%	-0.056%	-0.055%	0.001
SVN	549	-81%	-100%	-33%	-32%	-30%	-36%	-0.077%	-0.090%	-0.013
SWE	1,731	-74%	-100%	-31%	-26%	-46%	-58%	-0.023%	-0.031%	-0.007

Notes. This table extends Table 4 to the individual member states. The first column reports baseline exports of banned products to Russia in millions of USD. Export changes are relative to these baseline flows and include the markups paid to intermediaries. Real income changes are $(\hat{W} - 1) \times 100$. In the EU row, the export change aggregates member flows by value, while the real income change is the median member-state change, as in Table 4. The results are discussed in Section 6.